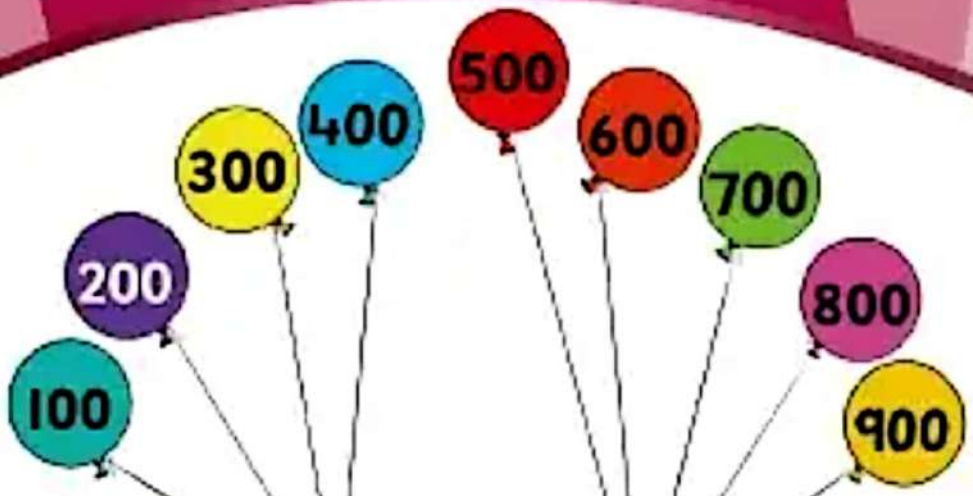
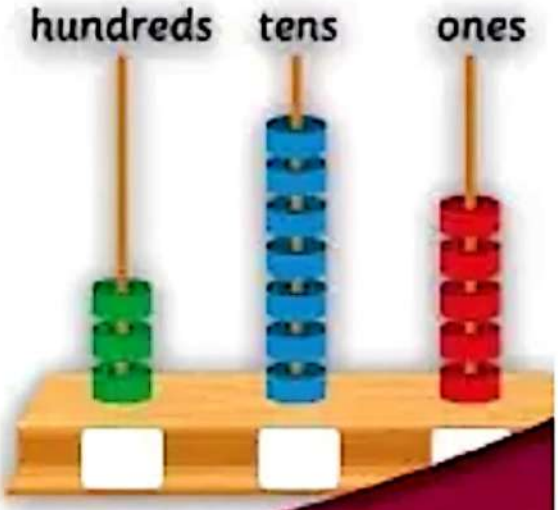
Arithmetic

Pupil's Book
Standard Two





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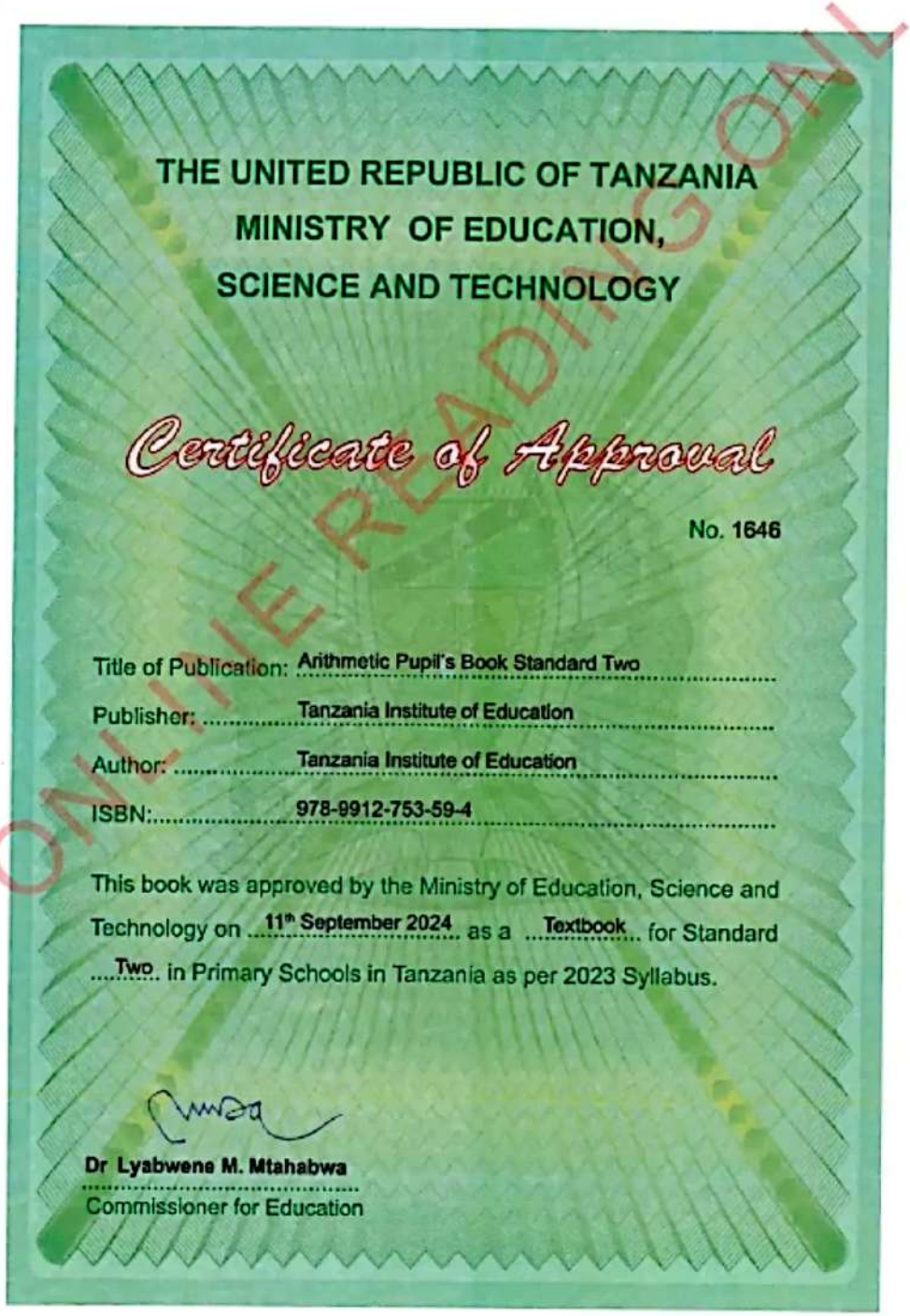

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Dr Lyabwene M. Mtahabwa
Commissioner for Education

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Dr Aneth A. Komba

Director General

Tanzania Institute of Education

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Introduction

This Arithmetic book has been prepared based on the 2023 Curriculum and Syllabus for Primary School Education issued by the Ministry of Education, Science and Technology. This book is the Second Edition of the Arithmetic book published in 2019 based on the 2016 curriculum. This book will enable the pupil to build arithmetic skills through various learning activities with emphasis in the following areas:

Mastering the basics of arithmetic

Mastering the skills of arithmetic

Applying relevant arithmetic principles

Additional learning resources are available at

TIE e-Library https://ol.tie.go.tz/videos

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Recognizing numbers in hundreds

Reading and writing numbers from 101 to 999

Reading and writing numbers are the basis of learning arithmetic. Here you will learn to read and write numbers not exceeding 999.

Let us sing a song.

Who has my lost bag? x 2.
Let us count numbers.
One hundred and one, one hundred and two, one hundred and three, one hundred and four, one hundred and five.
Who has my lost bag? x 2.
Let us count numbers.

One hundred and six, one hundred and seven, one hundred and eight, one hundred and nine, one hundred and ten.
Who has my lost bag? x 2.
Let us count numbers.

One hundred and eleven, one hundred and twelve, one hundred and thirteen, one hundred and fourteen, one hundred and fifteen.

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Exercise 1

Write in numerals all the numbers in the previous song.

Reading numbers from 101 to 999

Exercise 2

Read all the numbers from 101 to 150 in the following table.

101	102	103	104	105	106	107	108	109	110
111	112	113	114	115	116	117	118	119	120
121	122	123	124	125	126	127	128	129	130
131	132	133	134	135	136	137	138	139	140
141	142	143	144	145	146	147	148	149	150

Activity

Let us play a counting game

Let us stand in the form of a circle. Give each other enough distance apart to throw a ball. The first person to throw a ball should say 101. The person who catches the ball should say 102 when throwing it. The game continues until everyone is involved. This game is played by ten pupils standing and arranged in a circle. The following picture shows how to play this game.

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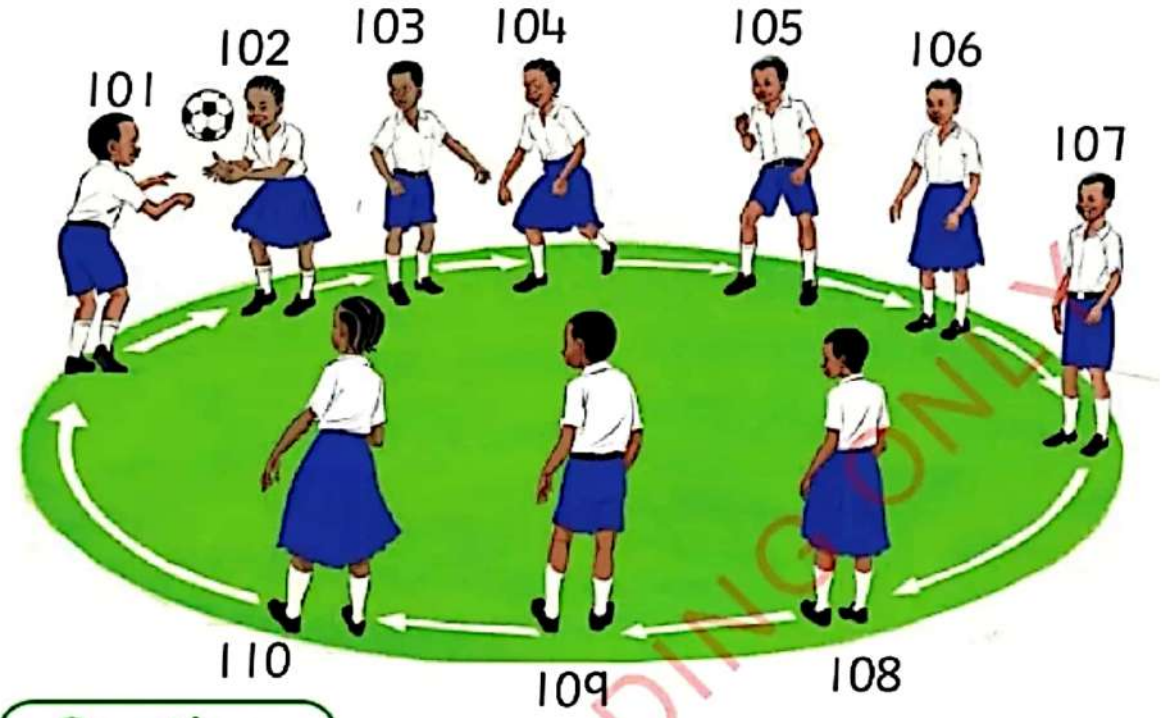

Question

Read all the numbers mentioned in the game.

Exercise 3

Read all the numbers from 151 to 200 in the following table.

151	152	153	154	155	156	157	158	159	160
161	162	163	164	165	166	167	168	169	170
171	172	173	174	175	176	177	178	179	180
181	182	183	184	185	186	187	188	189	190
191	192	193	194	195	196	197	198	199	200

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Exercise 4

Read all the numbers from 201 to 300 in the following table.

201	202	203	204	205	206	207	208	209	210
211	212	213	214	215	216	217	218	219	220
221	222	223	224	225	226	227	228	229	230
231	232	233	234	235	236	237	238	239	240
241	242	243	244	245	246	247	248	249	250
251	252	253	254	255	256	257	258	259	260
261	262	263	264	265	266	267	268	269	270
271	272	273	274	275	276	277	278	279	280
281	282	283	284	285	286	287	288	289	290
291	292	293	294	295	296	297	298	299	300

Exercise 5

Read all the numbers from 301 to 400 in the following table.

301	302	303	304	305	306	307	308	309	310
311	312	313	314	315	316	317	318	319	320
321	322	323	324	325	326	327	328	329	330
331	332	333	334	335	336	337	338	339	340
341	342	343	344	345	346	347	348	349	350
351	352	353	354	355	356	357	358	359	360
361	362	363	364	365	366	367	368	369	370
371	372	373	374	375	376	377	378	379	380
381	382	383	384	385	386	387	388	389	390
391	392	393	394	395	396	397	398	399	400

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Exercise 6

Read all the numbers from 401 to 500 in the following table.

401	402	403	404	405	406	407	408	409	410
411	412	413	414	415	416	417	418	419	420
421	422	423	424	425	426	427	428	429	430
431	432	433	434	435	436	437	438	439	440
441	442	443	444	445	446	447	448	449	450
451	452	453	454	455	456	457	458	459	460
461	462	463	464	465	466	467	468	469	470
471	472	473	474	475	476	477	478	479	480
481	482	483	484	485	486	487	488	489	490
491	492	493	494	495	496	497	498	499	500

Exercise 7

Read all the numbers from 501 to 665 in the following table.

501	502	503	504	505	506	507	508	509	510	511
512	513	514	515	516	517	518	519	520	521	522
523	524	525	526	527	528	529	530	531	532	533
534	535	536	537	538	539	540	541	542	543	544
545	546	547	548	549	550	551	552	553	554	555
556	557	558	559	560	561	562	563	564	565	566
567	568	569	570	571	572	573	574	575	576	577
578	579	580	581	582	583	584	585	586	587	588
589	590	591	592	593	594	595	596	597	598	599
600	601	602	603	604	605	606	607	608	609	610
611	612	613	614	615	616	617	618	619	620	621
622	623	624	625	626	627	628	629	630	631	632
633	634	635	636	637	638	639	640	641	642	643
644	645	646	647	648	649	650	651	652	653	654
655	656	657	658	659	660	661	662	663	664	665

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Exercise 8

Read all the numbers from 666 to 830 in the following table.

666	667	668	669	670	671	672	673	674	675	676
677	678	679	680	681	682	683	684	685	686	687
688	689	690	691	692	693	694	695	696	697	698
699	700	701	702	703	704	705	706	707	708	709
710	711	712	713	714	715	716	717	718	719	720
721	722	723	724	725	726	727	728	729	730	731
732	733	734	735	736	737	738	739	740	741	742
743	744	745	746	747	748	749	750	751	752	753
754	755	756	757	758	759	760	761	762	763	764
765	766	767	768	769	770	771	772	773	774	775
776	777	778	779	780	781	782	783	784	785	786
787	788	789	790	791	792	793	794	795	796	797
798	799	800	801	802	803	804	805	806	807	808
809	810	811	812	813	814	815	816	817	818	819
820	821	822	823	824	825	826	827	828	829	830

Exercise 9

Read all the numbers from 831 to 999 in the following table.

831	832	833	834	835	836	837	838	839	840	841
842	843	844	845	846	847	848	849	850	851	852
853	854	855	856	857	858	859	860	861	862	863
864	865	866	867	868	869	870	871	872	873	874
875	876	877	878	879	880	881	882	883	884	885

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886	887	888	889	890	891	892	893	894	895	896
897	898	899	900	901	902	903	904	905	906	907
908	909	910	911	912	913	914	915	916	917	918
919	920	921	922	923	924	925	926	927	928	929
930	931	932	933	934	935	936	937	938	939	940
941	942	943	944	945	946	947	948	949	950	951
952	953	954	955	956	957	958	959	960	961	962
963	964	965	966	967	968	969	970	971	972	973
974	975	976	977	978	979	980	981	982	983	984
985	986	987	988	989	990	991	992	993	994	995
996	997	998	999							

Exercise 10

Read the following numbers.

100	110	120	130	140	150	160	170	180	190
200	210	220	230	240	250	260	270	280	290
300	310	320	330	340	350	360	370	380	390
400	410	420	430	440	450	460	470	480	490
500	510	520	530	540	550	560	570	580	590
600	610	620	630	640	650	660	670	680	690
700	710	720	730	740	750	760	770	780	790
800	810	820	830	840	850	860	870	880	890
900	910	920	930	940	950	960	970	980	990

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Writing numbers from 101 to 999

Exercise 11

Write the following numbers in words.

1. 101 _____
2. 201 _____
3. 718 _____
4. 401 _____
5. 302 _____
6. 402 _____
7. 307 _____
8. 300 _____
9. 527 _____
10. 109 _____
11. 112 _____
12. 220 _____
13. 213 _____
14. 604 _____
15. 306 _____
16. 140 _____
17. 320 _____
18. 807 _____
19. 240 _____
20. 155 _____
21. 130 _____
22. 310 _____
23. 221 _____
24. 908 _____
25. 188 _____

Exercise 12

Write the following numbers in numerals:

1. One hundred and two _____
2. Two hundred and five _____
3. Five hundred _____
4. Four hundred and twenty-seven _____
5. Three hundred and three _____
6. Two hundred and twenty _____
7. One hundred and fifteen _____

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8. Four hundred and ninety _____
9. One hundred and nine _____
10. Three hundred and thirty _____
11. Two hundred and twenty-two _____
12. One hundred and one _____
13. Three hundred and fifty-five _____
14. Four hundred and one _____
15. Two hundred and twenty-five _____
16. Six hundred and one _____
17. Eight hundred and eight _____
18. Seven hundred and thirty-one _____
19. Nine hundred and ninety-nine _____
20. Eight hundred and thirteen _____

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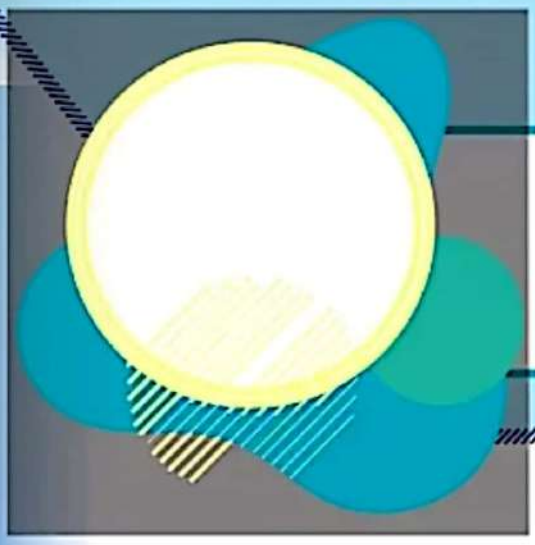

Arranging numbers sequentially

Arranging numbers sequentially from the smallest to the largest

Numbers can be arranged from the smallest to the largest as follows:

Example

Non-arranged numbers: 107 103 109 106 101

Arranged numbers: 101 103 106 107 109

Exercise 1

Arrange the following numbers sequentially from the smallest to the largest:

1.	245	237	220	253	212
	_____	_____	_____	_____	_____
2.	483	464	433	344	423
	_____	_____	_____	_____	_____
3.	377	368	381	399	309
	_____	_____	_____	_____	_____
4.	189	164	199	177	130
	_____	_____	_____	_____	_____

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Exercise 2

Arrange the following numbers sequentially from the smallest to the largest:

1.	240	450	499	311	301
2.	488	260	100	399	388

Exercise 3

Fill in the missing numbers in each of the following sequences:

1. 263, 264, _____, 266, _____
2. 109, 110, 111, _____, _____
3. 346, 347, _____, _____, 350
4. 488, 489, 490, _____, _____
5. _____, _____, 203, 204, 205

Exercise 4

Write the number that comes after in each of the following:

1.	409	_____	6.	419	_____
2.	467	_____	7.	350	_____
3.	259	_____	8.	262	_____
4.	199	_____	9.	319	_____
5.	206	_____	10.	375	_____

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Arranging numbers sequentially from the largest to the smallest

Numbers can be arranged sequentially from the largest to the smallest as follows:

Example

Non-arranged numbers:	146	113	169	188	199
Arranged numbers:	199	188	169	146	113

Exercise 5

Arrange the following numbers sequentially from the largest to the smallest:

1.	388	366	378	333	306
	_____	_____	_____	_____	_____
2.	159	138	127	118	144
	_____	_____	_____	_____	_____
3.	449	471	431	421	414
	_____	_____	_____	_____	_____
4.	266	269	278	297	255
	_____	_____	_____	_____	_____
5.	325	354	312	376	310
	_____	_____	_____	_____	_____

Exercise 6

Arrange the following numbers sequentially from the largest to the smallest:

1.	230	313	180	455	290
2.	129	440	310	500	282

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Exercise 7

Write the missing numbers in each of the following sequences:

1. 200, _____, 198
2. 451, _____, 449
3. 224, _____, 222
4. 380, _____, 378
5. 203, _____, 201
6. 315, _____, 113

Exercise 8

Write the number that comes before in each of the following sequences:

1. _____, 159, 158
2. _____, 313, 312
3. _____, 400, 399
4. _____, 210, 209
5. _____, 355, 354
6. _____, 407, 406
7. _____, 333, 332
8. _____, 179, 178
9. _____, 164, 163
10. _____, 220, 219

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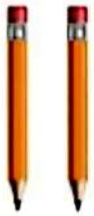
Counting objects

Counting objects in ones, tens, and hundreds

Objects can be counted in ones, tens, and hundreds.

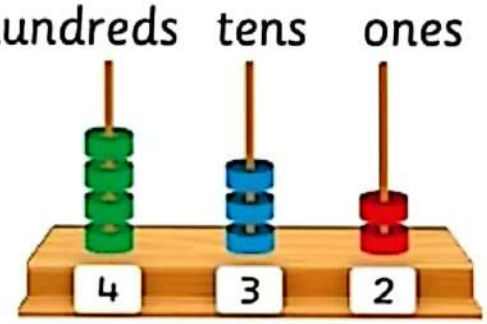
Example 1

Count the following pencils. Write their total in words.

Hundreds	Tens	Ones
		
One hundreds	Four tens	Two ones
One hundred and forty-two.		

Example 2

Count then write the number in words.



Four hundred and thirty-two.

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Example 3

Count the following pencils. Write their total in words.

Hundreds	Tens	Ones
One hundreds	One tens	Five ones
One hundred and fifteen.		

Example 4

Count the following pencils. Write their total in words.

Hundreds	Tens	Ones
One hundreds	One tens	One ones
One hundred and eleven.		

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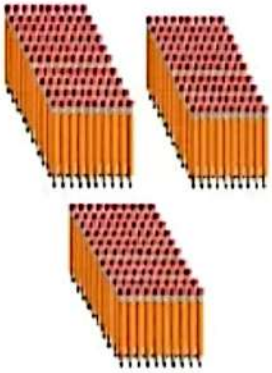
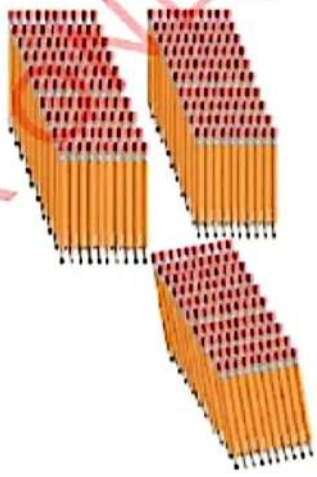
Example 5

Count the following pencils. Write their total in words.

Hundreds	Tens	Ones
		
Three hundreds	Two tens	Four ones
Three hundred and twenty-four.		

Exercise 1

Count the following pencils. Write their total in words.

1.	Hundreds	Tens	Ones
			
	_____	_____	_____
	_____		

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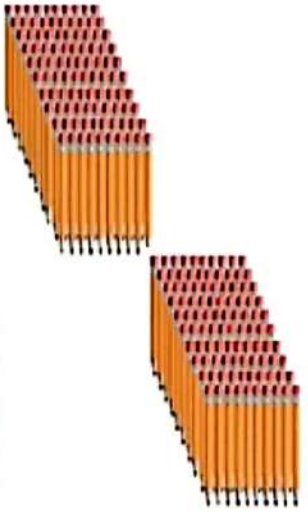
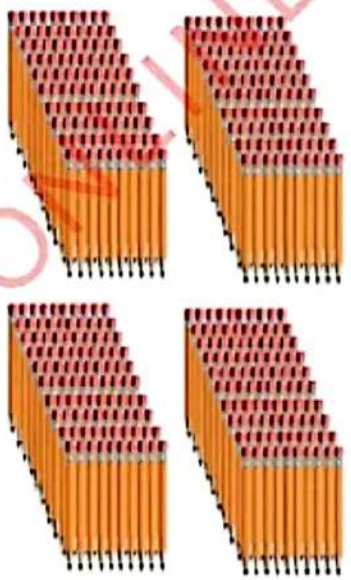
2.	Hundreds	Tens	Ones
			
	_____	_____	_____
_____			

3.	Hundreds	Tens	Ones
			
	_____	_____	_____
_____			

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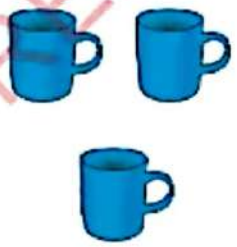
Exercise 2

Count the cups. Write their total in numerals in the blank space.

1.	Hundreds	Tens	Ones
	_____	_____	_____
2.	Hundreds	Tens 	Ones
	_____	_____	_____
3.	Hundreds	Tens	Ones
	_____	_____	_____

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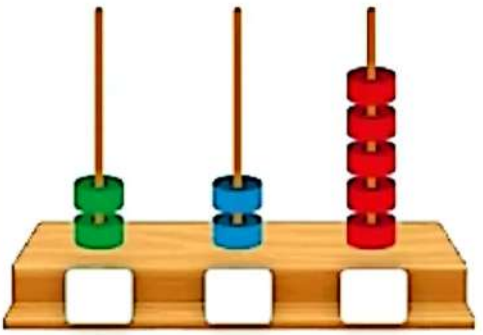
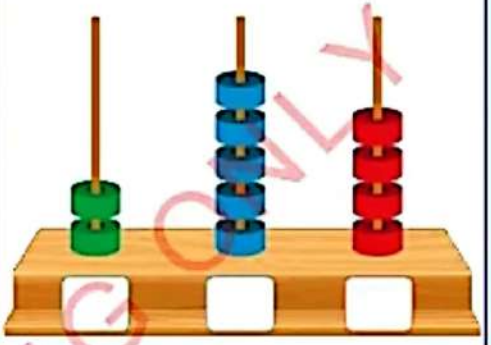
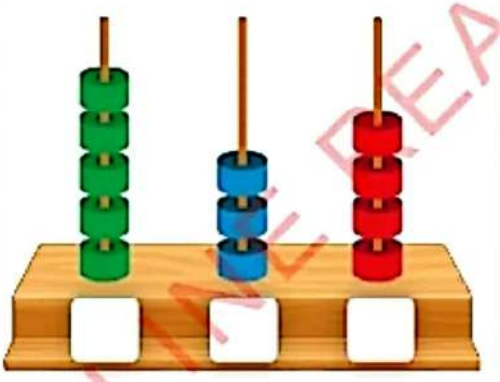
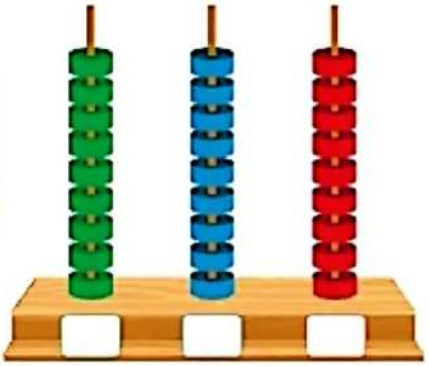
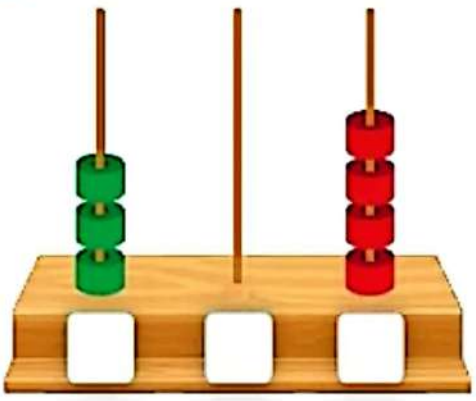
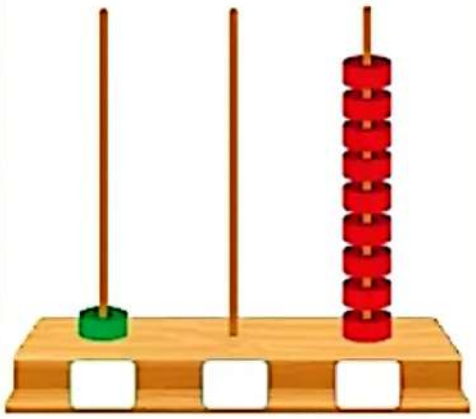
Exercise 3

Count then write the numbers in numerals and in words in the blank spaces.

1.	hundreds tens ones  _____	2.	hundreds tens ones  _____
3.	hundreds tens ones  _____	4.	hundreds tens ones  _____
5.	hundreds tens ones  _____	6.	hundreds tens ones  _____

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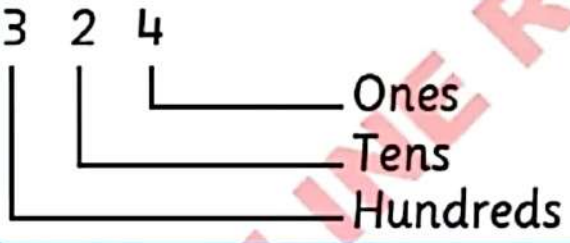
Place value of a digit

Recognizing the place value of a digit

Each position in a number carries a place value. The place value is recognized from the right towards the left of the number.

Example 1

Write the place value of each digit in 324.



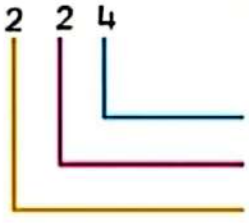
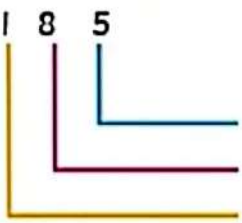
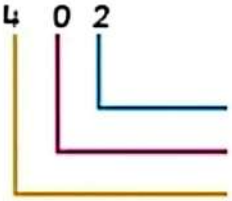
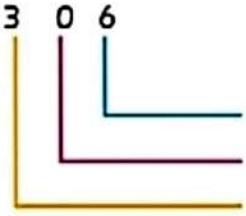
Example 2

Write the place value of each digit in 349.

349 = 3 hundreds 4 tens 9 ones

Exercise 1

Write the place value of each digit in the following:

1.		2.	
3.		4.	

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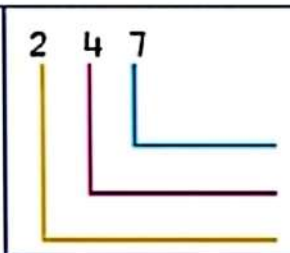
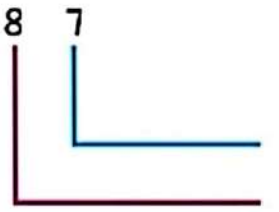
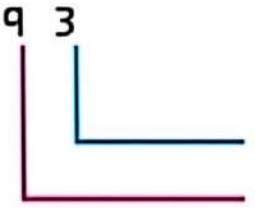
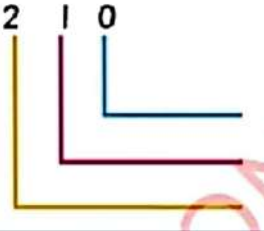
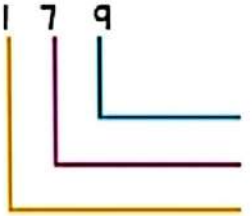
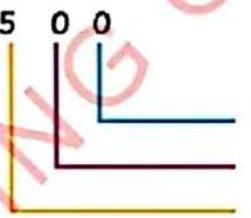
5.		6.	
7.		8.	
9.		10.	

Exercise 2

Write the digits that represent the place values of the following numbers:

1. 426 = ___ hundreds ___ tens ___ ones
2. 126 = ___ hundreds ___ tens ___ ones
3. 358 = ___ hundreds ___ tens ___ ones
4. 71 = ___ hundreds ___ tens ___ ones
5. 4 = ___ hundreds ___ tens ___ ones
6. 215 = ___ hundreds ___ tens ___ ones
7. 309 = ___ hundreds ___ tens ___ ones
8. 186 = ___ hundreds ___ tens ___ ones
9. 231 = ___ hundreds ___ tens ___ ones
10. 400 = ___ hundreds ___ tens ___ ones

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Exercise 3

Write the numbers which represent the following place values:

1.	2	hundreds	0	tens	1	ones	=	_____
2.	3	hundreds	3	tens	4	ones	=	_____
3.	0	hundreds	6	tens	9	ones	=	_____
4.	0	hundreds	0	tens	5	ones	=	_____
5.	4	hundreds	0	tens	0	ones	=	_____
6.	1	hundreds	2	tens	3	ones	=	_____
7.	2	hundreds	5	tens	6	ones	=	_____
8.	4	hundreds	1	tens	0	ones	=	_____
9.	3	hundreds	0	tens	8	ones	=	_____
10.	2	hundreds	4	tens	1	ones	=	_____

Writing the total value of a digit in a number

The total value of each digit in a number can be obtained as follows:

Example 1

Write the total value of each digit in 478.

4 7 8

Ones
Tens
Hundreds

The total value of 4 is 400.
The total value of 7 is 70.
The total value of 8 is 8.

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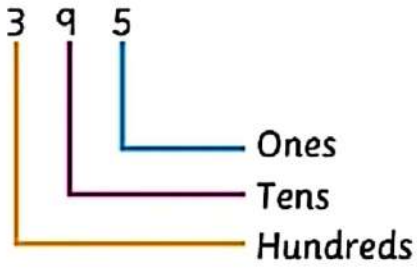
Example 2

Write the total value of each digit in 395.



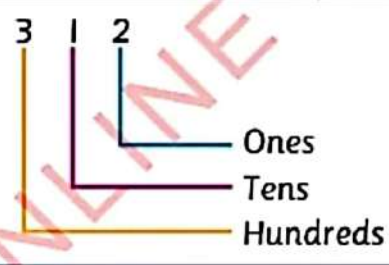
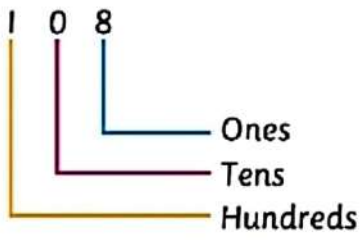
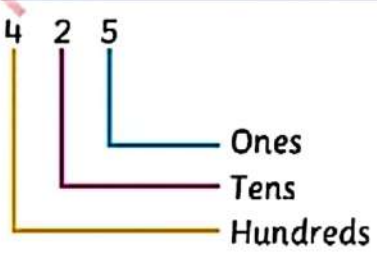
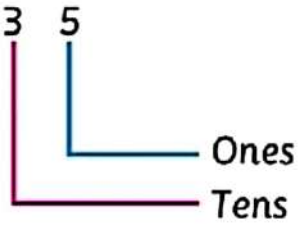
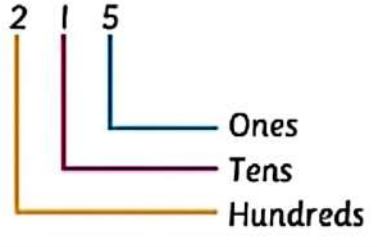
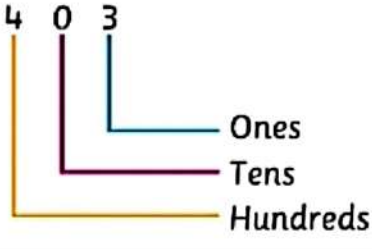
The total value of 3 is 300.

The total value of 9 is 90.

The total value of 5 is 5.

Exercise 4

Write the total value of each digit in the following numbers:

1.		2.	
3.		4.	
5.		6.	

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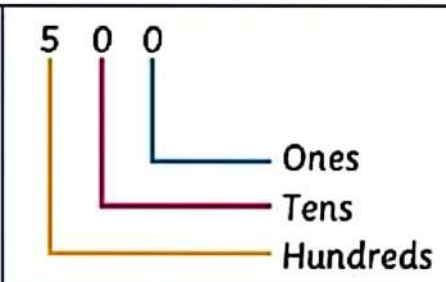
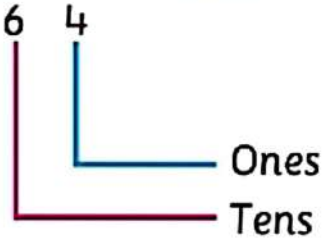
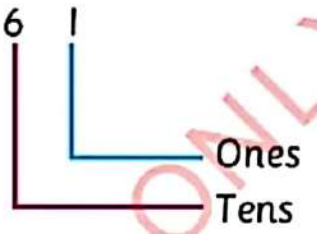
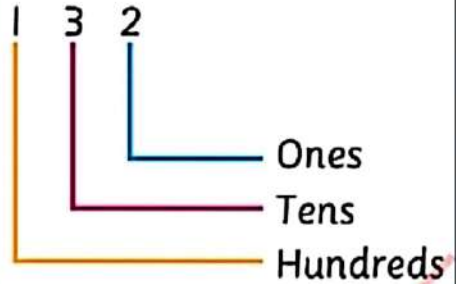
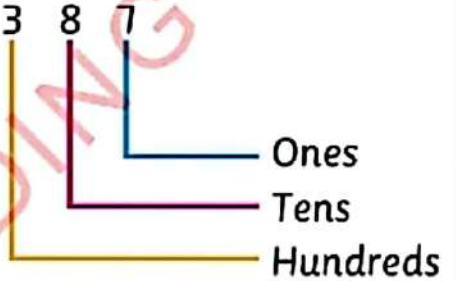
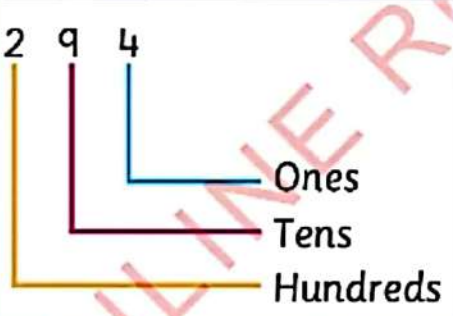
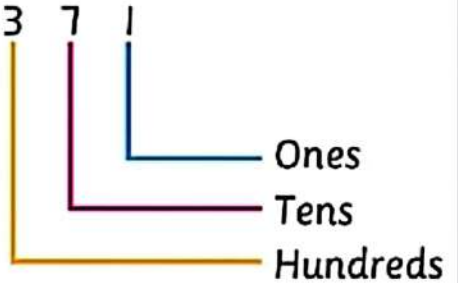
7.		8.	
9.		10.	
11.		12.	
13.		14.	

Exercise 5

Write the total value of each digit in the following numbers:

1. 201
2. 700
3. 333
4. 29
5. 218
6. 480
7. 11
8. 88
9. 999
10. 765

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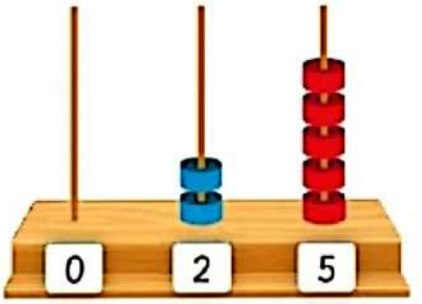
Exercise 6

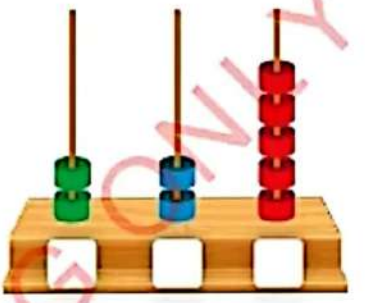
Write in numerals the numbers represented by the Abacus. Question 1 is an example.

1. hundreds tens ones



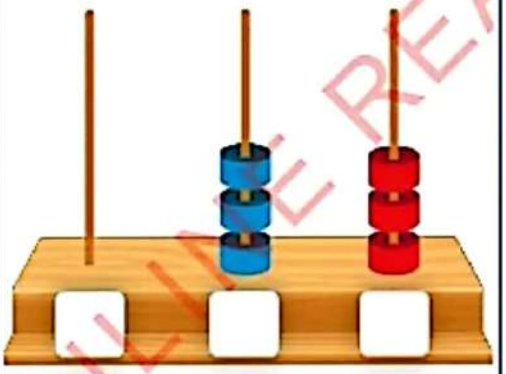
25

2. hundreds tens ones



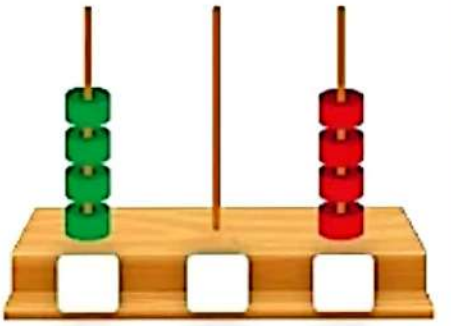
_____

3. hundreds tens ones



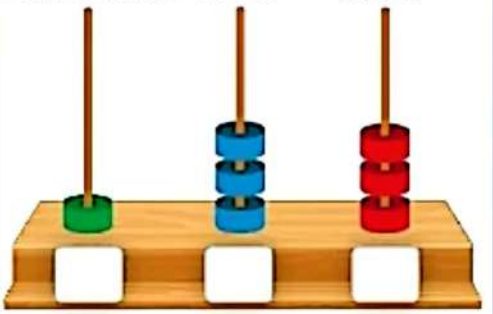
_____

4. hundreds tens ones



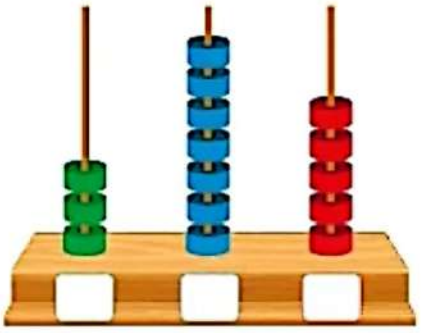
_____

5. hundreds tens ones



_____

6. hundreds tens ones



_____

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Writing numbers in expanded form

A number can be written in expanded form using the total value of its digits.

Example 1

Write 462 in expanded form.

Answer:

462 = 400 + 60 + 2.

Example 2

Write 109 in expanded form.

Answer:

109 = 100 + 0 + 9.

Exercise 7

Write the following numbers in expanded form.

1. 420 = _____
2. 46 = _____
3. 8 = _____
4. 302 = _____
5. 172 = _____
6. 93 = _____
7. 214 = _____
8. 163 = _____
9. 57 = _____
10. 291 = _____

Writing numbers in short form

A number in expanded form can be written in a short form as shown below.

Example 1

Write 200 + 70 + 3 in short form.

Answer:

200 + 70 + 3 = 273.

Example 2

Write 300 + 40 + 2 in short form.

Answer:

300 + 40 + 2 = 342.

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Exercise 8

Write the following numbers in short form:

1. 200 + 30 + 9 = _____
2. 100 + 20 + 7 = _____
3. 300 + 50 + 2 = _____
4. 400 + 80 + 6 = _____
5. 200 + 70 + 1 = _____
6. 100 + 40 + 3 = _____
7. 400 + 60 + 5 = _____
8. 300 + 10 + 8 = _____
9. 200 + 90 + 4 = _____
10. 300 + 30 + 3 = _____

Activity 1

Read the following passage and answer the questions that follow.

Manka has three children; Asha, Jane, and Hamisi. One day, she bought many exercise books and distributed them to her children. Asha got six exercise books. Jane got ten exercise books. Hamisi got two hundred exercise books as shown in the following picture.

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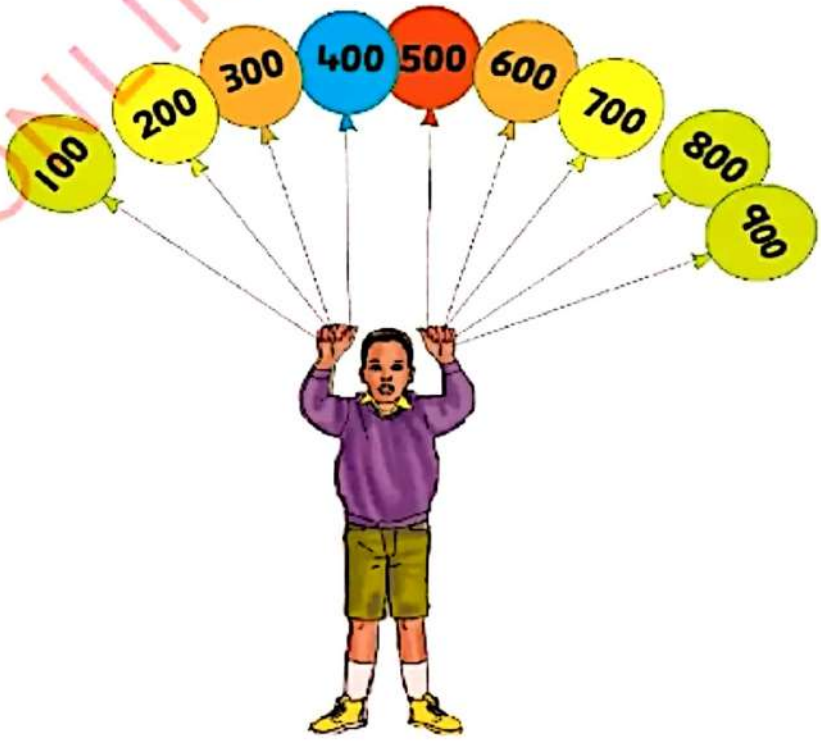
Questions

1. Write the number of exercise books given to Asha.
2. How many ones are there in the number of exercise books given to Asha?
3. Write the number of exercise books given to Jane.
4. How many tens are there in the number of exercise books given to Jane?
5. Write the number of exercise books given to Hamisi.
6. How many hundreds are there in the number of exercise books given to Hamisi?
7. How many exercise books did Manka distribute?
8. Write the place value of each digit in the number of exercise books distributed by Manka.

Activity 2

Study the following picture and answer the questions that follow.



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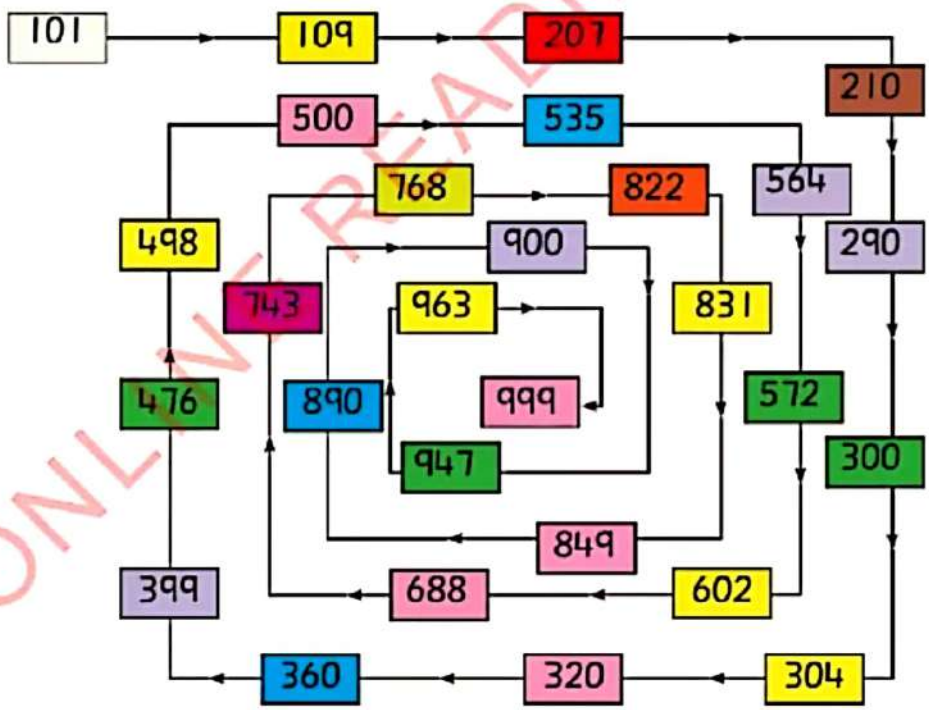
Questions

1. Write the number with the lowest hundreds.
2. Write the number with the largest hundreds.
3. Read the numbers shown in the picture from smallest to the largest.
4. Read the numbers shown in the picture from largest to the smallest.

Activity 3

Read the following numbers in the direction of the arrows. Answer the questions that follow.



Questions

1. Read loudly the ones of the first ten numbers.
2. Read loudly the hundreds of five numbers before 999.
3. Read loudly all the numbers ending with 0 ones.
4. Read loudly all the numbers ending with 0 tens.

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Exercise 9

1. Write the place value of each missing number horizontally in the following chart.

201	202	203		205	206	207	208	209
301		303	304	305	306	307	308	309
401	402	403	404	405		407	408	409
501	502	503	504		506	507	508	509
601		603	604	605	606	607	608	609
701	702	703	704	705	706		708	709
801	802	803	804	805	806	807	808	
901	902		904	905	906	907	908	909

2. Write the numbers with 9 ones in question 1.

3. Study the following table. Write digits that represent hundreds, tens, and ones in the blanks provided.

Number	Hundreds	Tens	Ones
229		2	
368	3		
876			6
569		6	
997	9		

4. Write the numbers formed by the place value of each of the following:

Hundreds	Tens	Ones	Number
1	1	9	_____
8	2	7	_____
2	6	3	_____
6	1	0	_____
3	3	3	_____

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Addition

Adding numbers not exceeding 999

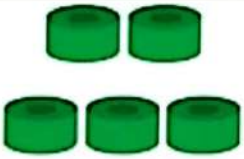
Numbers can be added horizontally or vertically. Addition is done by considering the place value of each digit in a number.

Adding numbers horizontally without regrouping

Numbers are added starting from the right to the left.

Example 1

223 + 345 =

Hundreds	Tens	Ones
Add	Add	Add
Equals	Equals	Equals
		
5	6	8
Therefore, 345 + 223 = 568.		

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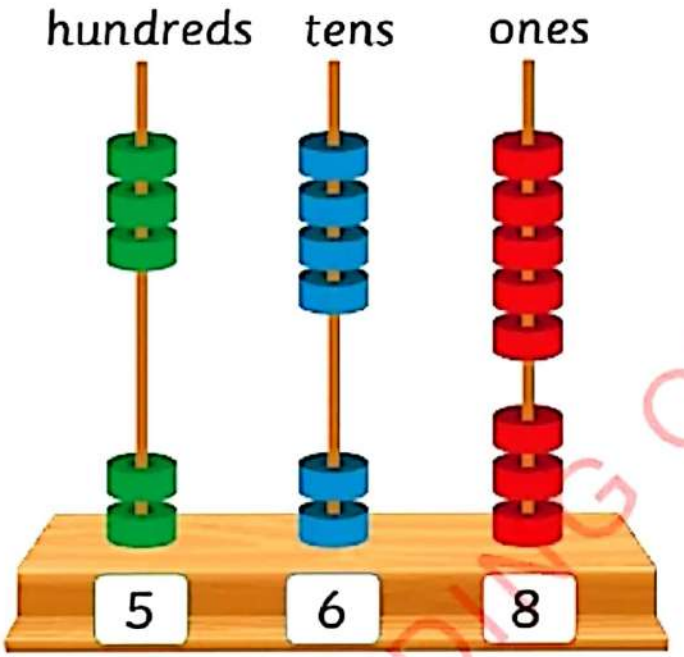
Or

223 + 345 =



Steps

1. Add ones: 3 + 5 = 8. Write 8 in the ones place.
   345 + 223 = 8.
2. Add tens: 2 + 4 = 6. Write 6 in the tens place.
   345 + 223 = 68.
3. Add hundreds: 2 + 5 = 5. Write 5 in the hundreds place.
   345 + 223 = 568.

Therefore, 345 + 223 = 568.

Example 2

512 + 415 =

Steps

1. Add ones: 2 + 5 = 7. Write 7 in the ones place.
   512 + 415 = 7
2. Add tens: 1 + 1 = 2. Write 2 in the tens place.
   512 + 415 = 27
3. Add hundreds: 5 + 4 = 9. Write 9 in the hundreds place.
   512 + 415 = 927

Therefore, 512 + 415 = 927

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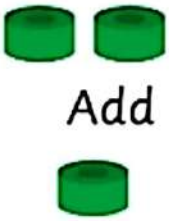
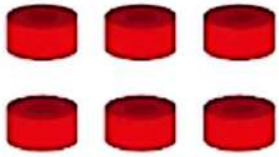
Exercise 1

1. Fill in the blanks in each of the following questions.

214 + 143 =

Hundreds	Tens	Ones
 Add	Add	Add
Equals	Equals	Equals
_____	_____	_____
___	___	___
_____		

2. 342 + 436 =

Hundreds	Tens	Ones
Add	Add	Add 
Equals	Equals	Equals
_____	_____	_____
___	___	___
_____		

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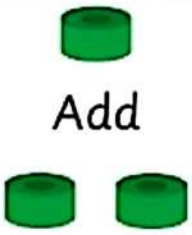
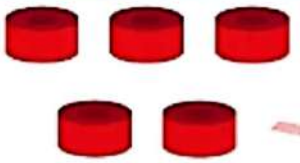
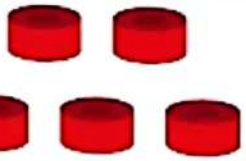
3. 125 + 233 =

Hundreds	Tens	Ones
 Add	Add	 Add
Equals	Equals	Equals
_____	_____	_____
___	___	___
_____		

4. 425 + 512 =

Hundreds	Tens	Ones
Add	Add	 Add
Equals	Equals	Equals
_____	_____	_____
___	___	___
_____		

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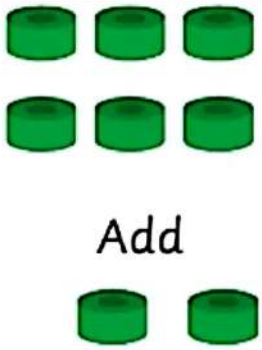
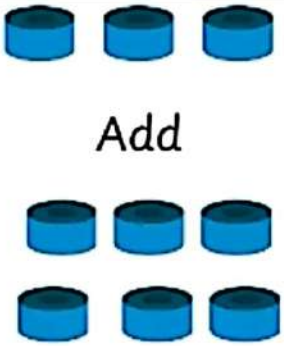
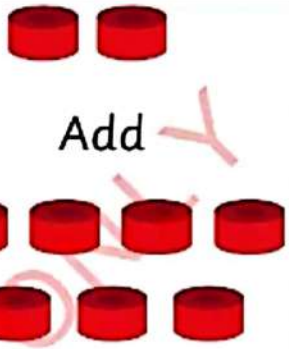
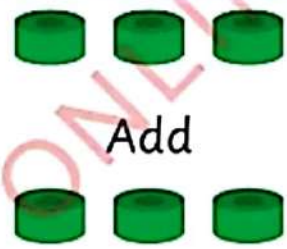
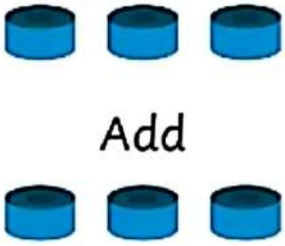
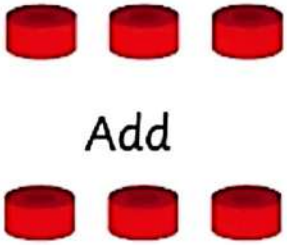
5. 632 + 267 =

Hundreds	Tens	Ones
 Add	 Add	 Add
Equals	Equals	Equals
_____	_____	_____
___	___	___
_____		

6. 333 + 333 =

Hundreds	Tens	Ones
 Add	 Add	 Add
Equals	Equals	Equals
_____	_____	_____
___	___	___
_____		

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Exercise 2

Write the answer in each question.

1. 111 + 333 =
2. 231 + 353 =
3. 213 + 534 =
4. 421 + 523 =
5. 345 + 534 =
6. 222 + 516 =
7. 245 + 723 =
8. 642 + 257 =
9. 800 + 100 =
10. 913 + 86 =
11. 666 + 222 =
12. 733 + 213 =
13. 444 + 123 =
14. 876 + 123 =
15. 586 + 413 =
16. 590 + 309 =
17. 478 + 221 =
18. 345 + 543 =
19. 654 + 324 =
20. 896 + 103 =

Activity 1

Read the following passage and answer the questions that follow:

Maria has three boxes for keeping money. In the first box, she kept one hundred shillings coins only. She kept only ten shillings coins in the second box. In the third box, she kept one shilling coins only as shown in the following picture.

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Box 1 Box 2 Box 3

Maria counted the number of coins in each box. She got nine coins of one hundred shillings, nine coins of ten shillings, and nine coins of one shilling.

Questions

1. How much money did Maria put in the first box?
2. How much money did Maria put in the first two boxes?
3. How much money did Maria put in all the three boxes?
4. Find the total amount of money in the first and third boxes.
5. Find the total amount of money in the last two boxes.

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Adding numbers vertically without regrouping

Numbers can be added vertically from right to the left.

Example 1

Add vertically.

Solution

	hundreds	tens	ones
2 1 6	2	1	6
+ 1 0 2	+ 1	0	2

Steps

1. Add ones: 6 + 2 = 8.
   Write 8 in the ones place.

	hundreds	tens	ones
	2	1	6
+	1	0	2
			8

2. Add tens: 1 + 0 = 1.
   Write 1 in the tens place.

	hundreds	tens	ones
	2	1	6
+	1	0	2
		1	8

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3. Add hundreds: 2 + 1 = 3.
Write 3 in the hundreds place.

	hundreds	tens	ones
	2	1	6
+	1	0	2
	3	1	8

Therefore, the answer is 318.

Example 2

Solution

	hundreds	tens	ones
	3	4	5
+	2	2	3
	5	6	8

345
+223
———
———

Steps

1. Add ones: 5 + 3 = 8.
Write 8 in the ones place.

	hundreds	tens	ones
	3	4	5
+	2	2	3
			8

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2. Add tens: 4 + 2 = 6.
Write 6 in the tens place.

	hundreds	tens	ones
	3	4	5
+	2	2	3
		6	8

3. Add hundreds: 3 + 2 = 5.
Write 5 in the hundred place.

	hundreds	tens	ones
	3	4	5
+	2	2	3
	5	6	8

Therefore, the answer is 568.

Exercise 3

Write the answer in each question.

1. $\begin{array}{r} 226 \\ +\,432 \\ \hline \\ \hline \end{array}$

2. $\begin{array}{r} 342 \\ +\,\ 22 \\ \hline \\ \hline \end{array}$

3. $\begin{array}{r} 432 \\ +\,\ 41 \\ \hline \\ \hline \end{array}$

4. $\begin{array}{r} 325 \\ +143 \\ \hline \\ \hline \end{array}$

5. $\begin{array}{r} 333 \\ +304 \\ \hline \\ \hline \end{array}$

6. $\begin{array}{r} 137 \\ +200 \\ \hline \\ \hline \end{array}$

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7. 444 + 222 = ____

8. 70 + 20 = ____

9. 106 + 450 = ____

10. 121 + 712 = ____

11. 215 + 421 = ____

12. 560 + 439 = ____

13. 126 + 563 = ____

14. 561 + 423 = ____

15. 716 + 231 = ____

Exercise 4

Write the answer in each question.

1. 465 + 334 = ____

2. 613 + 244 = ____

3. 472 + 324 = ____

4. 715 + 223 = ____

5. 678 + 210 = ____

6. 421 + 136 = ____

7. 714 + 133 = ____

8. 352 + 415 = ____

9. 526 + 313 = ____

10. 564 + 213 = ____

11. 215 + 421 = ____

12. 128 + 541 = ____

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13. $\begin{array}{r} 213 \\ +331 \\ \hline \end{array}$

14. $\begin{array}{r} 213 \\ +431 \\ \hline \end{array}$

15. $\begin{array}{r} 785 \\ +214 \\ \hline \end{array}$

Adding numbers horizontally by regrouping

Numbers can be added horizontally by regrouping. Regrouping is done by changing groups from lower place values to higher place values.

Example

167 + 138 =

Solution

167 + 138 = 305

Steps

1. Add ones: 7 + 8 = 15. Regroup 15 ones into 1 tens and 5 ones. Write 5 in the ones place. Take 1 tens to the tens place.	1 167 + 138 = _ _5
2. Add tens: 1 + 6 + 3 = 10. Regroup 10 tens into 1 hundreds and 0 tens. Write 0 in the tens place. Take 1 hundreds to the hundreds place.	11 167 + 138 = _05

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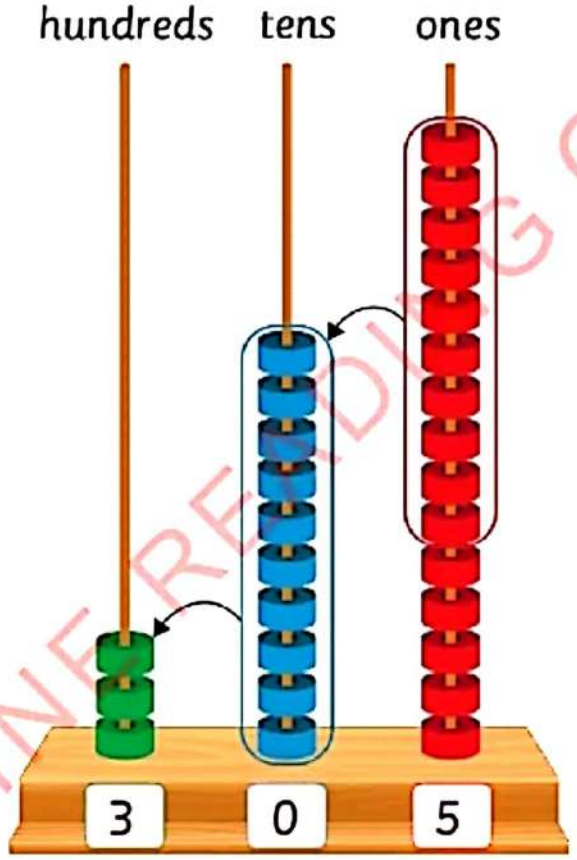
3. Add hundreds: 1 + 1 + 1 = 3. Write 3 in the hundreds place.	167 + 138 = 305
Therefore, the answer is 305.	

Or



Therefore, 167 + 138 = 305.

Exercise 5

Write the answer in each question.

1.	683 + 138 =	2.	364 + 348 =
3.	736 + 249 =	4.	262 + 289 =
5.	566 + 284 =	6.	292 + 362 =
7.	456 + 258 =	8.	272 + 465 =
9.	781 + 209 =	10.	189 + 771 =

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11.	484 + 109 =	12.	189 + 102 =
13.	801 + 9 =	14.	229 + 229 =
15.	518 + 119 =	16.	673 + 117 =
17.	239 + 221 =	18.	123 + 229 =
19.	444 + 137 =	20.	269 + 21 =

Exercise 6

Write the answer in each question.

1.	739 + 15 =	11.	291 + 19 =
2.	118 + 132 =	12.	839 + 111 =
3.	819 + 113 =	13.	709 + 29 =
4.	287 + 113 =	14.	469 + 31 =
5.	773 + 9 =	15.	729 + 148 =
6.	333 + 27 =	16.	456 + 326 =
7.	239 + 8 =	17.	381 + 409 =
8.	123 + 127 =	18.	462 + 418 =
9.	444 + 219 =	19.	282 + 365 =
10.	269 + 23 =	20.	627 + 203 =

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Adding numbers vertically by regrouping

Numbers can be added vertically by regrouping. Regrouping vertically is done in the same way as in horizontal addition.

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Example 1

Solution

	hundreds	tens	ones
166	1	6	6
+ 126	+ 1	2	6
	2	9	2

Steps

1. Add ones: 6 + 6 = 12. Regroup 12 ones into 1 tens and 2 ones. Write 2 in the ones place. Take 1 tens to the tens place.	1 166 +126 2
2. Add tens: 1 + 6 + 2 = 9. Write 9 in the tens place.	1 166 +126 92
3. Add hundreds: 1 + 1 = 2. Write 2 in the hundreds place.	1 166 +126 292
Therefore, the answer is 292.	

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Example 2

Solution

	hundreds	tens	ones
	1	6	6
+	1	9	8
	3	6	4

Steps

1. Add ones: 6 + 8 = 14. Regroup 14 ones into 1 tens and 4 ones. Write 4 in the ones place. Take 1 tens to the tens place.	1 1 6 6 + 1 9 8 4
2. Add tens: 1 + 6 + 9 = 16. Regroup 16 tens into 1 hundreds and 6 tens. Write 6 in the tens place. Take 1 hundreds to the hundreds place.	1 1 1 6 6 + 1 9 8 6 4
3. Add hundreds: 1 + 1 + 1 = 3. Write 3 to the hundreds place.	1 1 1 6 6 + 1 9 8 6 6 4
Therefore, the answer is 364.	

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Exercise 7

Write the answer in each question.

1. 435 + 219 = _____

2. 476 + 214 = _____

3. 665 + 192 = _____

4. 375 + 267 = _____

5. 295 + 494 = _____

6. 386 + 108 = _____

7. 150 + 252 = _____

8. 444 + 281 = _____

9. 568 + 372 = _____

10. 776 + 195 = _____

11. 329 + 295 = _____

12. 576 + 134 = _____

13. 99 + 1 = _____

14. 453 + 268 = _____

15. 336 + 485 = _____

Exercise 8

Write the answer in each question.

1. 576 + 234 = _____

2. 248 + 134 = _____

3. 356 + 275 = _____

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4. 486 + 116 = _____

5. 367 + 475 = _____

6. 237 + 295 = _____

7. 429 + 134 = _____

8. 586 + 236 = _____

9. 178 + 276 = _____

10. 285 + 37 = _____

11. 328 + 79 = _____

12. 467 + 295 = _____

13. 88 + 2 = _____

14. 179 + 95 = _____

15. 587 + 162 = _____

Adding numbers in a chart

The following chart has rows and columns of numbers.

+	101	102	103	104	105
101	202	203	204	205	206
102	203	204	205	206	207
103	204	205	206	207	208
104	205	206	207	208	209
105	206	207	208	209	210

Look at the first row. It has 101, 102, 103, 104 and 105. The first column has 101, 102, 103, 104 and 105. Observe where the rows and columns meet. Each number in the box is the sum of two numbers. Those numbers are in the first row and the first column.

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Exercise 9

Use the following chart to answer the questions.

+	201	202	203	204	205	206
201						
202						
203						
204					409	
205						
206			409			

1. Add the numbers in the chart.
2. Write the smallest number in the first row.
3. Write the largest number in the fifth row.
4. Write the largest number in the chart.
5. The largest number is the sum of which numbers?

Exercise 10

Fill in the missing numbers.

1. 102, 104, 106, ____, ____, ____
2. 105, 110, 115, ____, ____, ____
3. 200, 250, 300, ____, ____, ____
4. 820, 840, 860, ____, ____, ____
5. 890, 893, 896, ____, ____, ____
6. 799, 808, 817, ____, ____, ____
7. 601, 605, 609, ____, ____, 621, ____

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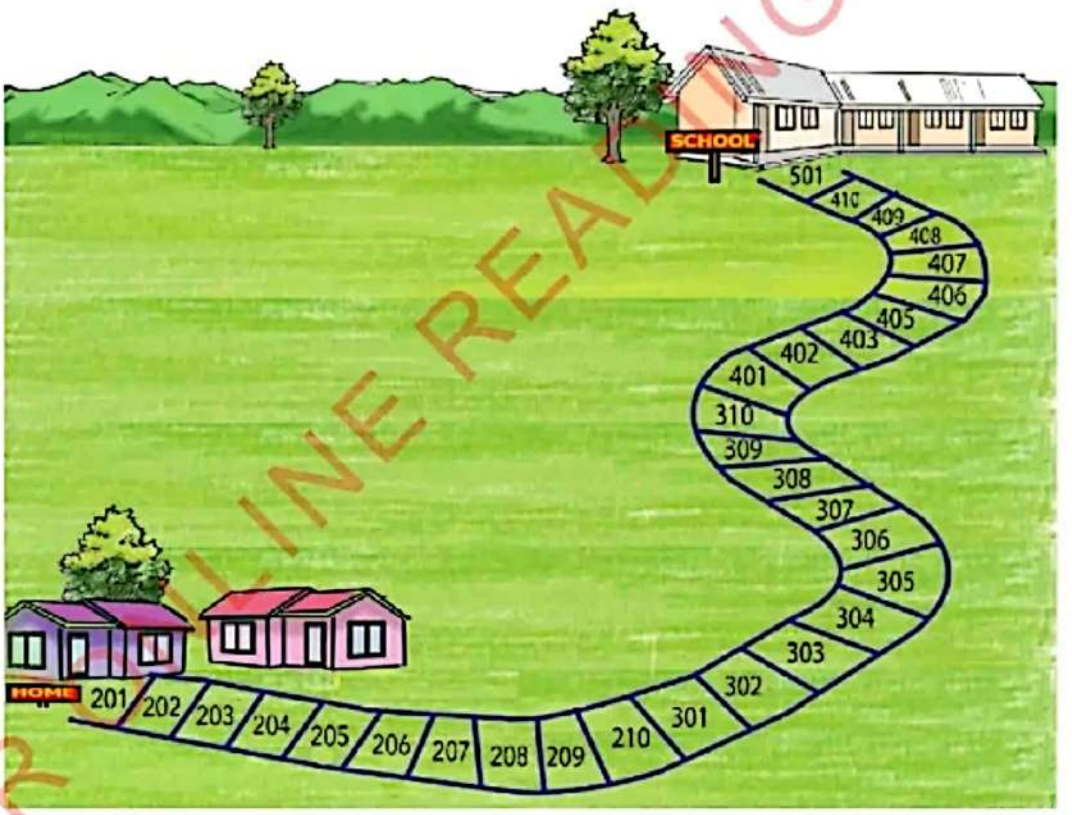
8. 401, 500, 599, ____, ____, ____
9. 306, 312, 318, ____, ____, ____
10. 497, 506, 515, ____, ____, 542, ____

Activity 2

Let us play a home-to-school game.

The game is played by using a die, a cup or a tin, and a marble. The following picture shows this game.



Steps

1. Play in pairs.
2. Use a cup or a tin.
3. Put a die in a cup or in a tin.
4. Each player must have a marble of different colour for identification.

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5. Roll the die in the cup or tin.
6. Count the number of dots shown on the top face of the die.
7. Move your marble in the direction of the school as the picture shows.
8. The marble is moved in steps equal to the number of dots on a die.
9. The first person to reach school will be the winner.

Example

Asha and Anna played a home-to-school game. Everyone played twice. Their scores were recorded as shown in the following table.

Game Number	Asha	Anna
1	203	205
2	206	211

The winner is the player who is closer to the school. Study the table and add the scores of each player.

Solution

Game Number	Asha	Anna
1	203	205
2	+ 206	+ 211
Total	409	416

The total scores of Asha is 409 and the total scores of Anna is 416.

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Questions

Use the home-to-school game to answer the following questions:

1. Name the tools used to play this game.
2. How many faces does a die have?
3. What is the largest number of dots on a die?
4. What is the smallest number of dots on a die?
5. What is the smallest number in the game?
6. What is the largest number in the game?
7. How many steps are there before the number 210 in the game?
8. Asha and Anna each rolled a die once. Both of their marbles were at 407. Asha got 2 dots and Anna got 4 dots. Who moved her marble up to number 501?

Word problems on addition

Word problems involving addition are arithmetic questions written in one or more sentence.

Example 1

A fisherman caught 112 fish in the morning. He then caught 210 fish in the afternoon. How many fish did he catch?

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Solution

Number of fish caught in the morning
Number of fish caught in the evening
Add the numbers vertically.

```
  1 1 2
+ 2 1 0
-------
  3 2 2
```

Steps

1. Add ones: 2 + 0 = 2. Write 2 in the ones place.	1 1 2 + 2 1 0 2
2. Add tens: 1 + 1 = 2. Write 2 in the tens place.	1 1 2 + 2 1 0 2 2
3. Add hundreds: 1 + 2 = 3. Write 3 in the hundreds place.	1 1 2 + 2 1 0 3 2 2
Therefore, a fisherman caught 322 fish.	

Example 2

A grandfather has 389 mango trees. A grandmother has 433 mango trees. How many mango trees do they both have?

Solution

Grandfather's mango trees = 389
Grandmother's mango trees = 433

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Add the numbers vertically.

$$\begin{array}{r} 389 \\ +433 \\ \hline 822 \\ \hline \end{array}$$

Steps

1.	Add ones: 9 + 3 = 12. Regroup 12 ones into 1 tens and 2 ones. Write 2 in the ones place. Take 1 tens to the tens place.	$\begin{array}{r} {\scriptstyle 1} \\ 389 \\ +\ 433 \\ \hline 22 \\ \hline \end{array}$
2.	Add tens: 1 + 8 + 3 = 12. Regroup 12 tens into 1 hundreds and 2 tens. Write 2 in the tens place. Take 1 hundreds to the hundreds place.	$\begin{array}{r} {\scriptstyle 1\ 1} \\ 389 \\ +\ 433 \\ \hline 22 \\ \hline \end{array}$
3.	Add hundreds: 1 + 3 + 4 = 8. Write 8 in the hundreds place.	$\begin{array}{r} {\scriptstyle 1\ 1} \\ 389 \\ +433 \\ \hline \\ \hline \end{array}$
Therefore, they have 822 mango trees.		

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Exercise 11

Answer the following questions.

1. One family had 130 goats. Another family had 110 goats. How many goats did the two families have?
2. A farmer planted 403 orange seedlings. She then planted 592 guava seedlings. How many seedlings did she plant altogether?
3. In a certain primary school, 372 pupils participated in sports. They were joined by 527 fellow pupils. How many pupils participated in the sports altogether?
4. A primary school bought 601 sports uniforms. Later, the school bought 326 more sports uniforms. How many sports uniforms did the school buy?
5. A training in environmental care was attended by 294 girls and 312 boys. How many people attended the training?
6. A school bought 333 desks. Later, the school bought another 426 desks. How many desks did the school buy?
7. Standard One pupils were given 350 pencils after passing an examination. Standard Two pupils were given 520 pencils after passing their examination. How many pencils were given to the two classes?

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8. A Head Teacher bought 150 exercise books. Later, she bought 216 more exercise books. How many exercise books did she buy altogether?
9. A village has 446 women and 358 men. How many people are there in the village?
10. A school has 590 pupils. Another school has 348 pupils. What is the total number of pupils in the two schools?
11. On Monday, 247 cows were sent to the open market. On Tuesday, 264 cows were sent to the open market. How many cows were sent to the open market in the two days?
12. A shopkeeper bought 516 eggs on the first day. He bought another 378 eggs on the second day. How many eggs did he buy in the two days?
13. A school shop had 464 bottles of soda. The shopkeeper added 385 more bottles of soda in the shop. How many bottles of soda are in the shop?
14. A primary school received a gift of 625 pencils. Later, it received 286 more pencils. How many pencils did the school get in total?
15. A health officer gave 766 mosquito nets to the villagers. Later, she gave 186 more mosquito nets to the villagers. How many mosquito nets did the villagers get?
16. One lorry carried 187 pawpaws. Another lorry carried 216 pawpaws. How many pawpaws did the two lorries carry?

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Subtraction

Subtracting numbers not exceeding 999

Numbers can be subtracted horizontally or vertically. Subtraction is done by considering the place value of each digit in a number.

Subtracting numbers horizontally without regrouping

Numbers are subtracted starting from the right to the left.

Example

247 – 123 =

Solution

hundreds	tens	ones		hundreds	tens	ones		hundreds	tens	ones
2	4	7	–	1	2	3	=	1	2	4

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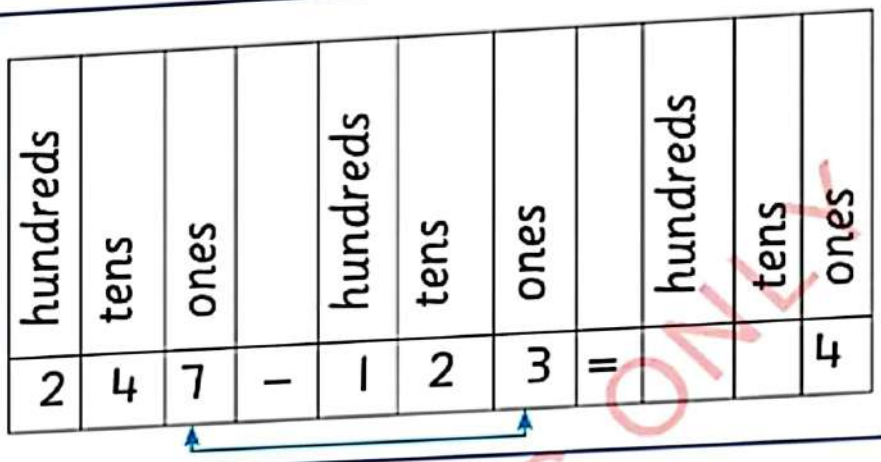
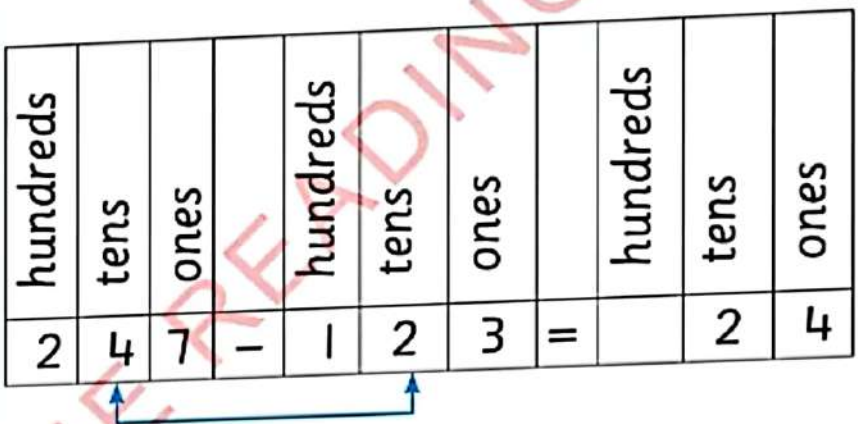
Steps

1. Subtract ones: $7 - 3 = 4$. Write 4 in the ones place.



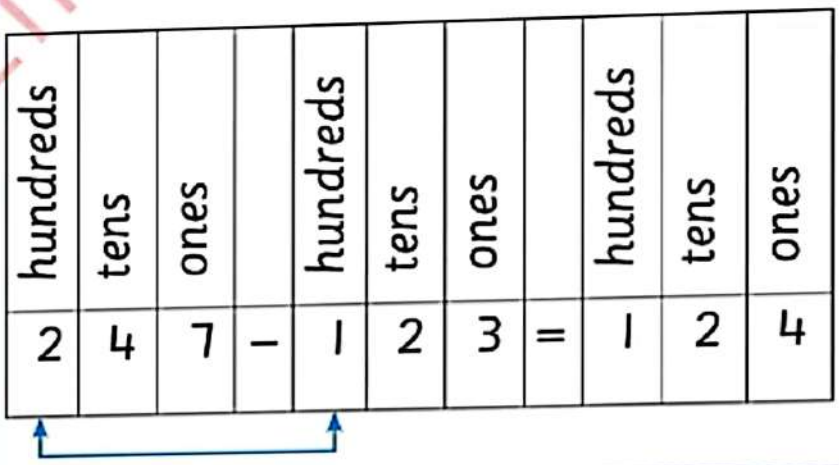
hundreds	tens	ones		hundreds	tens	ones		hundreds	tens	ones
2	4	7	−	1	2	3	=			4

2. Subtract tens: $4 - 2 = 2$. Write 2 in the tens place.



hundreds	tens	ones		hundreds	tens	ones		hundreds	tens	ones
2	4	7	−	1	2	3	=		2	4

3. Subtract hundreds: $2 - 1 = 1$. Write 1 in the hundreds place.



hundreds	tens	ones		hundreds	tens	ones		hundreds	tens	ones
2	4	7	−	1	2	3	=	1	2	4

Therefore, the answer is 124.

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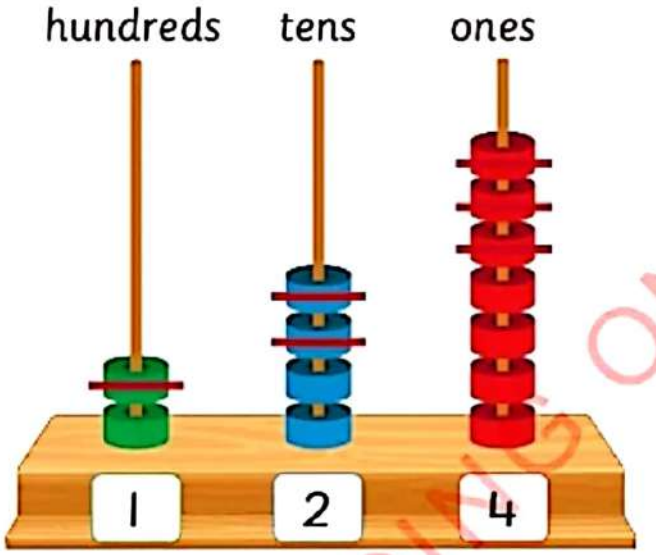
Or

247 – 123 =



Therefore, 247 – 123 = 124

Exercise 1

Write the answer in each question.

1. 299 – 133 =
2. 795 – 275 =
3. 444 – 200 =
4. 568 – 346 =
5. 306 – 204 =
6. 597 – 50 =
7. 980 – 440 =
8. 600 – 300 =
9. 649 – 48 =
10. 937 – 415 =
11. 935 – 213 =
12. 528 – 417 =
13. 438 – 216 =
14. 287 – 240 =
15. 386 – 215 =
16. 756 – 524 =
17. 368 – 118 =
18. 999 – 439 =

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Exercise 2

Write the answer in each question.

1. 436 – 321 =
2. 758 – 643 =
3. 239 – 115 =
4. 384 – 243 =
5. 895 – 670 =
6. 267 – 145 =
7. 530 – 410 =
8. 682 – 381 =
9. 999 – 717 =
10. 246 – 123 =
11. 896 – 643 =
12. 900 – 500 =
13. 729 – 408 =
14. 256 – 113 =
15. 446 – 325 =
16. 185 – 172 =
17. 388 – 216 =
18. 888 – 283 =

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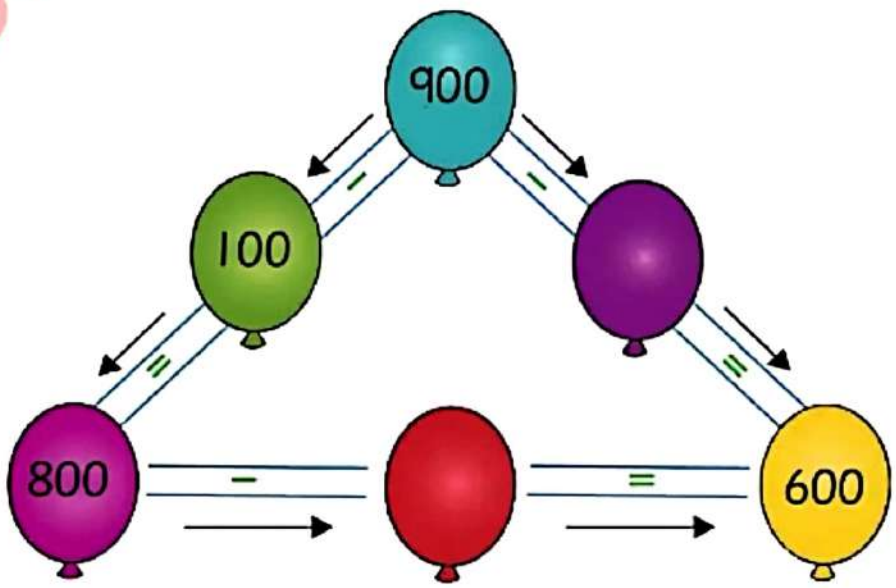
Activity 1

Subtraction game

1. Use the direction of the arrows to fill the missing numbers in the balloons.



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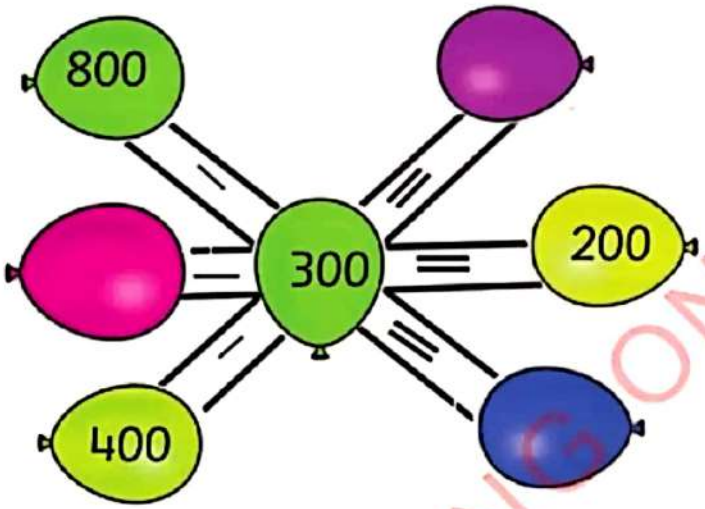
2. Fill in the missing numbers in the balloons.



Subtracting numbers vertically without regrouping

Numbers are subtracted vertically from the right to the left.

Example

Solution

$$\begin{array}{r} 668 \\ -346 \\ \hline \end{array} \qquad \begin{array}{r} 668 \\ -346 \\ \hline 322 \end{array}$$

Steps

1.	Subtract ones: 8 – 6 = 2. Write 2 in the ones place.	$\begin{array}{r} 668 \\ -346 \\ \hline 2 \end{array}$
2.	Subtract tens: 6 – 4 = 2. Write 2 in the tens place.	$\begin{array}{r} 668 \\ -346 \\ \hline 22 \end{array}$

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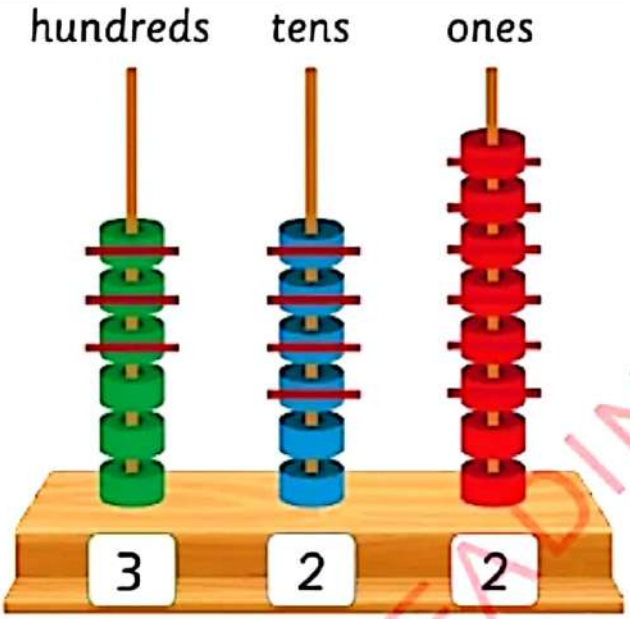
3.	Subtract hundreds: 6 – 3 = 3. Write 3 in the hundreds place.	6 6 8 – 3 4 6 3 2 2

Or



Therefore, the answer is 322.

Exercise 3

Write the answer in each question.

1. 1 7 8 − 1 7 5 = _____

2. 4 8 6 − 2 4 4 = _____

3. 4 9 0 − 2 5 0 = _____

4. 4 7 8 − 4 7 0 = _____

5. 4 4 5 − 3 3 4 = _____

6. 3 8 4 − 2 5 1 = _____

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7. 287
185

8. 357
−242

9. 248
−101

Exercise 4

Write the answer in each question.

1. 815
−604

2. 744
−333

3. 509
−104

4. 787
−645

5. 812
−710

6. 555
−324

7. 975
−230

8. 659
−548

9. 758
−234

Exercise 5

Write the answer in each question.

1. 632
−321

2. 799
−566

3. 431
−221

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4. 328 − 215 = _____

5. 776 − 715 = _____

6. 824 − 221 = _____

7. 624 − 321 = _____

8. 515 − 102 = _____

9. 345 − 215 = _____

467 − 115 = _____

349 − 117 = _____

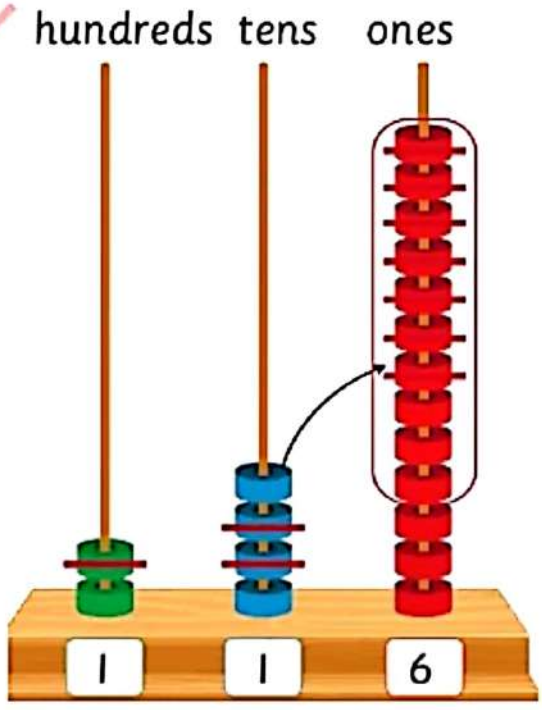
12. 286 − 123 = _____

Subtracting numbers horizontally by regrouping

Numbers can be subtracted horizontally by regrouping.

Example 1

243 − 127 =



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Steps

1. Put the beads in the abacus: 3 beads in the ones place, 4 beads in the tens place, and 2 beads in the hundreds place.
2. Subtract 7 beads from the ones place. It is not sufficient. There are only 3 beads in the ones place.
3. Take 1 group of tens from 4 tens. Regroup it into 10 ones. Remember that 3 tens remain.
4. Add ones: 10 + 3 = 13.
5. Subtract ones: 13 – 7 = 6. Write 6 in the ones place.
6. Subtract tens: 3 – 2 = 1. Write 1 in the tens place.
7. Subtract hundreds: 2 – 1 = 1. Write 1 in the hundreds place.

   Therefore, 243 – 127 = 116.

Example 2

100 – 47 =

Solution

100 – 47 = 53

Steps

1.	Subtract ones: 0 –7, it is not sufficient.	100 – 47 = □□

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2.	Take 1 group of tens from 0 tens. It is not sufficient. Take 1 group of hundreds from 1 hundreds. Regroup it into 10 tens. Remember that 0 hundreds remained in the hundreds place.	010 1Ø0 − 47 = ☐☐
3.	Take 1 tens from 10 tens. Regroup 1 tens into 10 ones. Remember 9 tens remained in the tens place.	0 9 10 1ØØ − 47 = ☐☐
4.	Subtract ones: 10 − 7 = 3. Write 3 in the ones place.	0 9 10 1ØØ − 47 = ☐3
5.	Subtract tens: 9 − 4 = 5. Write 5 in the tens place.	0 9 10 1ØØ − 47 = 53
6.	Subtract hundreds: 0 − 0 = 0. Leave a blank space in the hundreds place.	
	Therefore, the answer is 53.	

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Exercise 6

Write the answer in each question.

1. 823 – 147 =
2. 295 – 76 =
3. 864 – 576 =
4. 890 – 698 =
5. 100 – 21 =
6. 640 – 128 =
7. 584 – 286 =
8. 770 – 493 =
9. 586 – 138 =
10. 654 – 337 =
11. 778 – 469 =
12. 984 – 427 =
13. 881 – 457 =
14. 456 – 257 =
15. 551 – 499 =
16. 100 – 25 =

Exercise 7

Write the answer in each question.

1. 764 – 147 =
2. 354 – 215 =
3. 582 – 396 =
4. 755 – 267 =
5. 633 – 278 =
6. 987 – 498 =
7. 287 – 97 =
8. 384 – 295 =
9. 863 – 289 =
10. 692 -188 =
11. 449 – 187 =
12. 249 – 194 =
13. 904 – 106 =
14. 816 – 798 =
15. 387 – 197 =
16. 100 – 12 =

Subtracting numbers vertically by regrouping

Numbers can be subtracted vertically by regrouping.

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Example

	hundreds	tens	ones
3 1 4	3	1	4
−2 0 8	2	0	8
	1	0	6

Solution

Steps

1.	Subtract one: 4 − 8, it is not sufficient.	3 1 4 −2 0 8
2.	Take 1 group of tens from 1 tens. Regroup it into 10 ones. Remember 0 tens remain. Add ones: 10 + 4 = 14.	0 14 3 ~~1~~ ~~4~~ −2 0 8
3.	Subtract ones: 14 − 8 = 6. Write 6 in the ones place.	0 14 3 ~~1~~ ~~4~~ −2 0 8 6
4.	Subtract tens: 0 − 0 = 0. Write 0 in the tens place.	0 14 3 ~~1~~ ~~4~~ −2 0 8 0 6
5.	Subtract hundreds: 3 − 2 = 1. Write 1 in the hundreds place.	0 14 3 1 4 −2 0 8 1 0 6
	Therefore, the answer is 106.	

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Exercise 8

Write the answer in each question.

1. 564 − 235 = ____

2. 487 − 9 = ____

3. 600 − 482 = ____

4. 925 − 688 = ____

5. 841 − 759 = ____

6. 100 − 90 = ____

7. 753 − 65 = ____

8. 896 − 618 = ____

9. 856 − 67 = ____

10. 982 − 593 = ____

11. 896 − 617 = ____

12. 477 − 289 = ____

13. 878 − 498 = ____

14. 900 − 798 = ____

15. 100 − 19 = ____

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Exercise 9

Write the answer in each question.

1. 975 − 689 = _____

2. 647 − 429 = _____

3. 968 − 589 = _____

4. 548 − 299 = _____

5. 798 − 349 = _____

6. 784 − 495 = _____

7. 663 − 487 = _____

8. 817 − 709 = _____

9. 576 − 268 = _____

10. 754 − 269 = _____

11. 876 − 595 = _____

12. 582 − 397 = _____

13. 976 − 489 = _____

14. 801 − 779 = _____

15. 862 − 279 = _____

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Word problems on subtraction

Word problems involving subtraction are questions written in one or more sentence.

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Example 1

Kabula had 956 sweets. She gave her sibling 813 sweets. How many sweets did she remain with?

Solution

Number of sweets = 956

Number of sweets given away = 813

Subtract: 956 – 813 = 143.

Steps

1. Subtract ones: 6 – 3 = 3.
   Write 3 in the ones place. 956 – 813 = 3.
2. Subtract tens: 5 – 1 = 4.
   Write 4 in the tens place. 956 – 813 = 43.
3. Subtract hundreds: 9 – 8 = 1. Write 1 in the hundreds place. 956 – 813 = 143
   Therefore, Kabula remained with 143 sweets.

Example 2

A school had 543 desks. If 129 desks were damaged, how many desks were not damaged?

Solution

Total number of desks = 543

Number of damaged desks = 129

Subtract:

$$\begin{array}{r} {}^{3}\cancel{5}\,{}^{13}\cancel{4}\,\cancel{3} \\ -\ 1\ 2\ 9 \\ \hline 4\ 1\ 4 \end{array}$$

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Steps

1.	Subtract ones: 3 – 9, it is not sufficient.	5 4 3 – 1 2 9 _____
2.	Take 1 group of tens from 4 tens. Regroup 1 tens into 10 ones. Remember that 3 tens remained in the tens place. Add ones: 10 + 3 = 13.	3 13 5 4 ~~3~~ – 1 2 9 4
3.	Subtract ones: 13 – 9 = 4. Write 4 in the ones place.	3 13 5 4 ~~3~~ – 1 2 9 4
4.	Subtract tens: 3 – 2 = 1.Write 1 in the tens place.	3 13 5 4 ~~3~~ – 1 2 9 1 4
5.	Subtract hundreds: 5 – 1 = 4. Write 4 in the hundreds place.	3 13 5 4 ~~3~~ – 1 2 9 4 1 4
Therefore, 414 desks were not damaged.		

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Exercise 10

Answer the following word problems.

1. Shija had 649 eggs. If he sold 415 eggs, how many eggs were left?
2. A passenger train carried 874 pupils. On the way, 360 pupils got off the train. Find the number of

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pupils who continued with the journey.

3. A teacher had 596 books. He gave out 421 books to his pupils. How many books did the teacher remain with?
4. Roza had 365 oranges. If she sold 365 oranges, how many oranges did she remain with?
5. There were 635 pencils in a shop. If 412 pencils were sold, how many pencils were left in the shop?
6. A second-hand toy vendor had 888 toys. He sold 573 toys. Find the number of toys that remained.
7. A businesswoman bought 760 bottles of juice. She sold 650 bottles. How many bottles was she left with?
8. Erika had 500 shillings for buying exercise book. If an exercise book costs 355 shillings, how much money did she remain with?
9. Some ducks laid 734 eggs. If 218 eggs were broken, how many eggs remained?
10. A flock of 500 birds landed on a tree. Some 308 birds flew away. How many birds remained on the tree?
11. Juma had 463 oranges. He sold 185 oranges. How many oranges remained?
12. In a class of 210 students, 170 students sat for an examination. How many students did not sit for the examination?

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Multiplication

Multiplication using objects

Multiplication is a repeated addition of a number or objects. The operation of multiplication is represented by the sign ×. For example, 2 × 1 is read as, 2 multiplied by 1. It is also read as 2 times 1.

Example 1

2 × 1 =

Steps

1. Put two cups in one group only once. This means, 2 × 1

2. Add the number of cups. The sum is 2 cups. Therefore, 2 × 1 = 2.

Example 2

2 × 2 =

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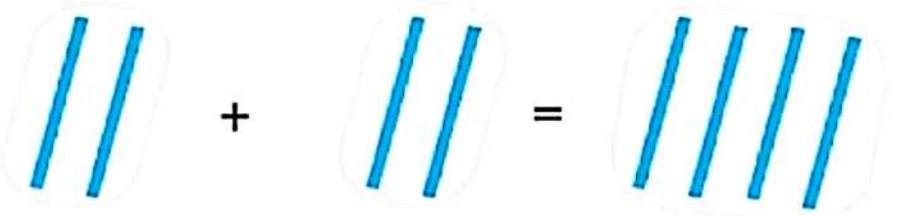
Solution



Two sticks + Two sticks = Four sticks

Therefore, $2 \times 2 = 4$.

Example 3

$3 \times 2 =$

Solution

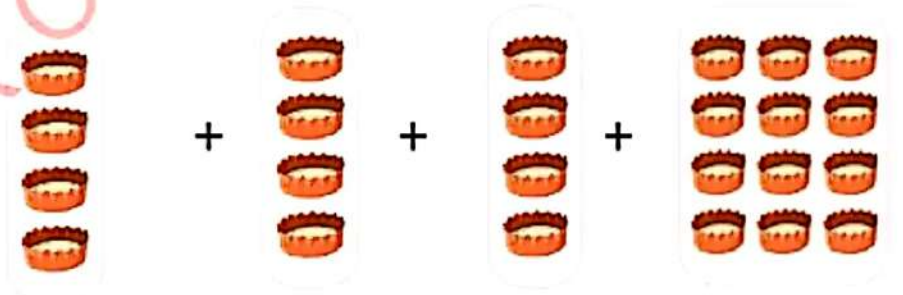
Three balls + Three balls = Six balls

Therefore, $3 \times 2 = 6$.

Example 4

$3 \times 2 =$

Solution



Four bottle tops + Four bottle tops + Four bottle tops + Twelve bottle tops

Therefore, $4 \times 2 = 12$

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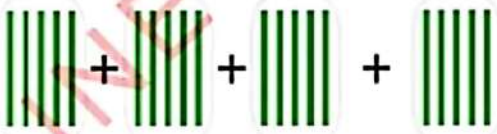
Exercise 1

Use objects to multiply each of the following numbers.

1.	4 × 2		=	___ cars
2.	6 × 3		=	___ pencils
3.	_____		=	___ bells
4.	_____		=	___ sticks

Multiplying using repeated addition

Multiplication can be done using repeated addition of a number.

Example 1

2 × 4 =

Solution

Repeated addition: 2 + 2 + 2 + 2 = 8.

Multiply: 2 × 4 = 8.

Therefore, 2 × 4 = 8.

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Example 2

$4 \times 3 =$

Solution

Repeated addition: $4 + 4 + 4 = 12$.

Multiply: $4 \times 3 = 12$

Therefore, $4 \times 3 = 12$.

Example 3

$3 \times 5 =$

Solution

Repeated addition: $3 + 3 + 3 + 3 + 3 = 15$.

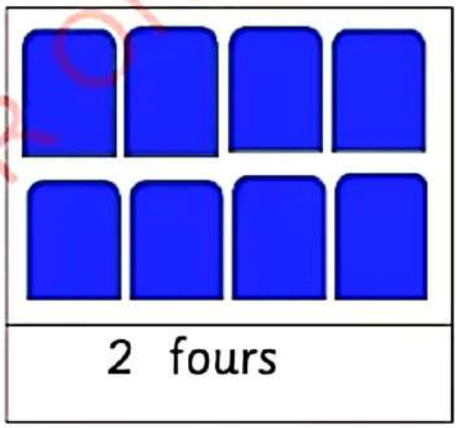
Multiply: $3 \times 5 = 15$

Therefore, $3 \times 5 = 15$.

Example 4

4×2

Solution



Adding: $4 + 4 = 8$.

$4 \times 2 = 8$

Therefore, $4 \times 2 = 8$.

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Exercise 2

Fill in the blank boxes. Question 1 is an example.

1.	2 × 6 =	2 + 2 + 2 + 2 + 2 + 2 =	12
2.	4 × 7 =	4 + 4 + 4 + 4 + 4 + 4 + 4 =	
3.	4 × 5 =	☐ =	
4.	☐ =	6 + 6 + 6 + 6 =	
5.	☐ =	5 + 5 + 5 + 5 + 5 =	25
6.	8 × 3 =	☐ =	
7.	10 × 2 =	☐ =	
8.	9 × 5 =	☐ =	
9.	11 × 4 =	☐ =	
10.	12 × 3 =	☐ =	

Exercise 3

Write the answer in each question.

1. 2 × 1 =
2. 2 × 2 =
3. 2 × 3 =
4. 5 × 4 =
5. 2 × 5 =
6. 6 × 3 =
7. 7 × 2 =
8. 2 × 8 =
9. 2 × 9 =
10. 10 × 4 =
11. 2 × 11 =
12. 2 × 12 =
13. 9 × 2 =
14. 8 × 4 =

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Exercise 4

Fill in the missing numbers in the following table.

×	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5				
2	2	4	6	8	10				
3	3	6	9	12	15				
4									
5									
6									
7									
8									
9									

Multiplying numbers horizontally

A two digit number can be multiplied by a one digit number horizontally.

Example 1

$24 \times 2 =$

Steps

1. Multiply 4 ones by 2: $4 \times 2 = 8$. Write 8 in the ones place.
2. Multiply 2 tens by 2: $2 \times 2 = 4$. Write 4 in the tens place.

Therefore, $24 \times 2 = 48$.

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Example 2

$4 \times 21 =$

Steps

1. Multiply 4 by 1 ones: 1: $4 \times 1 = 4$. Write 4 in the ones place.
2. Multiply 4 by 2 tens: 2: $4 \times 2 = 8$. Write 8 in the tens place.

Therefore, $4 \times 21 = 84$.

Exercise 5

Write the answer in each question.

1. $13 \times 2 =$
2. $11 \times 7 =$
3. $14 \times 2 =$
4. $10 \times 5 =$
5. $30 \times 2 =$
6. $33 \times 3 =$
7. $12 \times 3 =$
8. $11 \times 2 =$
9. $10 \times 4 =$
10. $3 \times 13 =$
11. $5 \times 11 =$
12. $4 \times 12 =$
13. $12 \times 2 =$
14. $2 \times 40 =$

Multiplying numbers vertically

A two digit number can be multiplied by a one digit number vertically.

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Example 1

$$\begin{array}{r} 13 \\ \times\ 2 \\ \hline \\ \hline \end{array}$$

Steps

1. Multiply 3 ones by 2: $3 \times 2 = 6$.
   Write 6 in the ones place.
2. Multiply 1 tens by 2: $1 \times 2 = 2$.
   Write 2 in the tens place.
   Therefore, the answer is 26.

Example 2

$$\begin{array}{r} 34 \\ \times\ 2 \\ \hline \\ \hline \end{array}$$

Steps

1. Multiply 4 ones by 2: $4 \times 2 = 8$.
   write 8 in the ones place.
2. Multiply 3 tens by 2: $3 \times 2 = 6$.
   Write 6 in the tens place.
   Therefore, the answer is 68.

Exercise 6

Write the answer in each question.

1. $\begin{array}{r} 21 \\ \times\ 3 \\ \hline \\ \hline \end{array}$

2. $\begin{array}{r} 32 \\ \times\ 3 \\ \hline \\ \hline \end{array}$

3. $\begin{array}{r} 22 \\ \times\ 4 \\ \hline \\ \hline \end{array}$

4. $\begin{array}{r} 16 \\ \times\ 1 \\ \hline \\ \hline \end{array}$

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5. $\begin{array}{r} 33 \\ \times\ 3 \\ \hline \\ \hline \end{array}$

6. $\begin{array}{r} 40 \\ \times\ 2 \\ \hline \\ \hline \end{array}$

7. $\begin{array}{r} 14 \\ \times\ 2 \\ \hline \\ \hline \end{array}$

8. $\begin{array}{r} 12 \\ \times\ 4 \\ \hline \\ \hline \end{array}$

9. $\begin{array}{r} 44 \\ \times\ 2 \\ \hline \\ \hline \end{array}$

10. $\begin{array}{r} 11 \\ \times\ 9 \\ \hline \\ \hline \end{array}$

Multiplying numbers by zero

Zero means there is nothing. When a number is multiplied by 0, the answer is 0. For example, you want to pick 2 oranges from the garden but there are no oranges. This means you will not have any orange. This is written $0 \times 2 = 0$. If you go 2 times and comeback without an orange, you will not have any orange. This is written $0 \times 2 = 0$.

Example 1

$0 \times 2 = 0$.

Example 2

$0 \times 2 =$

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Word problems on multiplication

Word problems involving multiplication are questions written in one or more sentence.

Example 1

There are 6 flower beds in a garden. Every flower bed has 5 seedlings. How many seedlings are there in the garden?

Solution

Number of flower beds = 6.

Number of seedlings in each flower bed = 5.

Multiply the numbers:

$6 \times 5 = 6 + 6 + 6 + 6 + 6 = 30$.

Therefore, the garden has 30 flower seedlings.

Example 2

One car can carry 22 passengers at once. How many passengers can be carried by 3 cars of the same capacity?

Solution

Number of passengers in one car = 22

Number of cars = 3

Multiply:

$$\begin{array}{r} 22 \\ \times\ 3 \\ \hline 66 \end{array}$$

Therefore, 3 cars can carry 66 passengers.

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Exercise 8

Answer the following questions:

1. A class has 34 pupils. How many pupils are there in 2 similar classes?
2. A school received iron sheets from 8 businessmen. 3. Every businessman gave 5 iron sheets. How many iron sheets did the school get?
4. An automotive industry manufactures 12 cars every month. Find the number of cars manufactured in three months?
5. A carpenter makes 3 tables every day. How many tables does he make in 30 days?
6. The traffic department trains 44 drivers on road safety measures every month. How many drivers are trained in 2 months?
7. A teacher bought 2 boxes of chalks. Each box contained 30 pieces of chalk. How many pieces of chalk did the teacher buy?
8. A teacher taught a lesson on HIV/AIDS to standards I and II pupils. If each class had 30 pupils, how many pupils were taught the lesson on HIV/AIDS?
9. A farmer has 2 dairy cows. Each cow produces 5 litres of milk a day. How many litres of milk does the farmer get in a day?
10. A teacher gave an equal number of biscuits to 6 pupils. Each pupil got 4 biscuits. How many biscuits did the teacher give?

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10. John planted 4 rows of orange seedlings in his garden. He planted 12 seedlings in each row. How many seedlings did he plant?

Multiplying numbers by using ICT

Numbers can be multiplied by using ICT.

Example

1 × 5 =

Solution

Use the computer application in the TIE online library https:// tie.go.tz/pages/download-software to multiply numbers.

Steps

1. Open the application for multiplication of numbers.
2. Read the question that appears on the screen clearly.
3. Type the answer by clicking on the keyboard.
4. Click OK to check your answer.
5. Do the next questions to practice multiplication of numbers by using ICT.

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Division

Division using objects

Division is the process of distributing objects into groups of equal number of objects. It is the opposite of multiplication. It is also known as repeated subtraction. The operation of division is represented by the sign ÷. For example, 8 ÷ 4 = 2 is read as, 8 divided by 4 equals 2.

When dividing objects, it is better to distribute one object at a time, in rounds.

Example

Divide 6 stars into 2 groups. How many stars will each group get?

Steps

1. Share one star in each group at a time as follows:

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First share

Group one	Group two

Second share

Group one	Group two

Third share

Group one	Group two

2. Count the number of stars in each group after the third share. The answer is the number of stars in each group.

Therefore, each group will get 3 stars.

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Activity

Use counters to perform the following activities:

1. Distribute 30 counters into 6 equal groups. How many counters are there in each group?
2. Distribute 36 counters into 4 equal groups. How many counters are there in each group?
3. Distribute 96 counters into 3 equal groups. How many counters are there in each group?

Dividing objects by using numbers

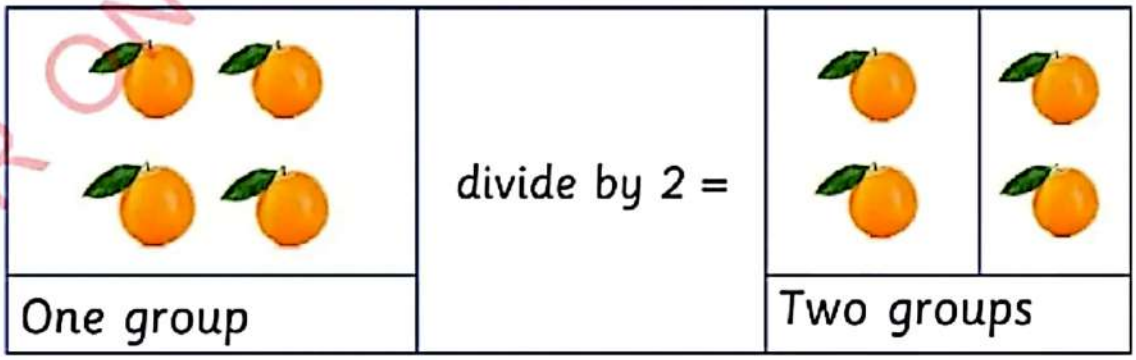
Example 1

Divide equally 4 oranges to 2 children. How many oranges will each child receive?

Solution

Dividing 4 oranges to 2 children can be written as,

$4 \div 2 =$



Thus, $4 \div 2 = 2$.

Therefore, each child will get 2 oranges.

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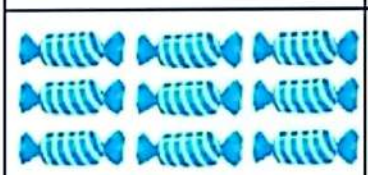
Example 2

Kajuna, Masanja, and Zuwena shared 9 sweets equally. How many sweets did each one get?

Solution

9 ÷ 3 =

Total sweets		Kajuna	Masanja	Zuwena
	divide by 3 =			
One group		Three groups		

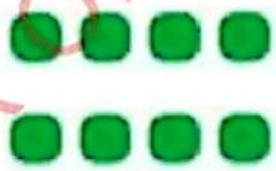
Thus, 9 ÷ 3 = 3.

Therefore, each one will get 3 sweets.

Example 3

Divide 8 objects by 4.

Solution

 One group	divide by 4 =	Four groups

Thus, 8 ÷ 4 = 2.

Therefore, there are four groups with 2 objects each.

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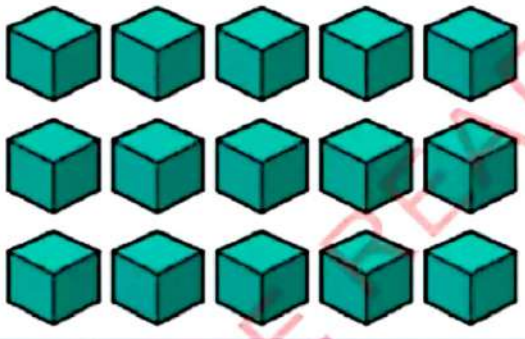
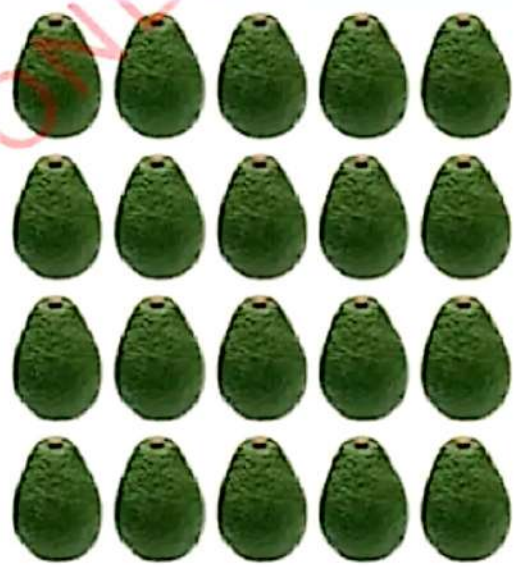
Exercise 1

Study the following pictures. Divide the given numbers.

1.		18 ÷ 3 =
2.		15 ÷ 5 =
3.		6 ÷ 3 =
4.		20 ÷ 5 =

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5.		$16 \div 4 =$

Division by using repeated subtraction

Repeated subtraction is the process of subtracting the same number repeatedly. The process stops when a 0 is obtained.

Example 1

$6 \div 2 =$

Solution

Subtract 2 from 6 repeatedly.

$6 - 2 = 4$

$6 - 2 = 2$

$6 - 2 = 0$

Repeated subtraction has been done 3 times.

Therefore, $6 \div 2 = 3$

Example 2

$8 \div 4 =$

Solution

Subtract 4 from 8 repeatedly.

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$8 - 4 = 4$

$4 - 4 = 0$

Repeated subtraction has been done 2 times.
Therefore, $8 \div 4 = 2$

Exercise 2

Use repeated subtraction to answer the following questions:

1. $4 \div 2 =$
2. $10 \div 2 =$
3. $24 \div 2 =$
4. $12 \div 3 =$
5. $20 \div 5 =$
6. $16 \div 4 =$
7. $16 \div 8 =$
8. $15 \div 3 =$
9. $24 \div 6 =$
10. $9 \div 3 =$
11. $81 \div 9 =$
12. $18 \div 6 =$
13. $15 \div 5 =$
14. $48 \div 4 =$
15. $8 \div 2 =$

Exercise 3

Write the answer in each question.

1. $84 \div 7 =$
2. $42 \div 7 =$
3. $28 \div 7 =$
4. $64 \div 4 =$
5. $25 \div 5 =$
6. $63 \div 9 =$
7. $22 \div 2 =$
8. $36 \div 9 =$
9. $32 \div 4 =$
10. $32 \div 8 =$
11. $72 \div 9 =$
12. $99 \div 3 =$
13. $88 \div 8 =$
14. $48 \div 6 =$
15. $66 \div 6 =$

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Word problems on division

Word problems involving division are questions written in one or more sentence.

Example 1

A story book has 12 pages. Dezi can read 3 pages of the book everyday. How long will she take to read the whole book?

Solution

Divide 12 pages by 3.

$12 \div 3 = 4$

Therefore, Dezi will take 4 days to read the whole book.

Example 2

A group of 6 pupils shared 36 mangoes equally. How many mangoes did each pupil get?

Solution

Divide 36 by 6.

$36 \div 6 = 6$

Therefore, each pupil got 6 mangoes.

Exercise 4

Answer the following questions:

1. Fourteen guavas were divided into 7 equal groups. How many guavas will be in each group?

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2. Share equally 20 buns to 5 friends. How many buns will each friend get?
3. A bunch of 30 carrots were equally distributed in 10 small bunches. How many carrots does each bunch have?
4. A total of 42 exercise books were equally shared to 7 pupils. How many exercise books did each pupil get?
5. A total of 32 loaves of bread were equally distributed to 8 people. How many loaves of bread did each person get?
6. A total of 45 pencils were equally distributed to 5 pupils. How many pencils did each pupil get?
7. A tourist gave a gift of 64 rubbers to 8 pupils. If they equally shared them, how many rubbers did each pupil get?
8. A total of 72 eggs were equally laid by 6 chickens. How many eggs did each chicken lay?
9. A total of 40 books were equally distributed to 4 pupils. How many books did each pupil get?
10. One hundred hectares of land were distributed to 10 villagers. How many hectors did each villager get?

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Exercise 5

Read the following paragraph and then answer the questions that follow.

A teacher had several gifts for 6 pupils. The gifts were; 18 pencils, 12 rubbers, 30 apples and some biscuits.

1. The teacher distributed the pencils to the pupils equally. How many pencils did each pupil get?
2. The teacher distributed 60 biscuits equally. How many biscuits did each pupil get?
3. The teacher distributed the apples to the pupils equally. How many apples did each pupil get?
4. The teacher distributed rubbers to each pupil equally. How many rubbers did each pupil get?
5. The number of pupils was increased to 9. How many pencils did each pupil get?

Dividing numbers by using ICT

Numbers can be divided by using ICT.

Example

$7 \div 1 =$

Solution

Use the computer application in the TIE online library https:// tie.go.tz/pages/download-software to divide numbers.

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Steps

1. Open the application for division of numbers.
2. Read the question that appears on the screen clearly.
3. Type the answer by clicking on the keyboard.
4. Click OK to check your answer.
5. Do other questions to practice division of numbers.

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Recognizing fractions

Recognizing one-half of a whole object

Two equal parts of the same object form one whole object. A half is one of the two equal parts. For example, an orange can be cut into two equal pieces. Each piece is one half of the orange. Formally, one-half is also known as, a half.

Example 1

Whole object	First half	Second half
an orange	half an orange	half an orange

Example 2

Whole object	First half	Second half
a pineapple	half a pineapple	half a pineapple

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Example 3

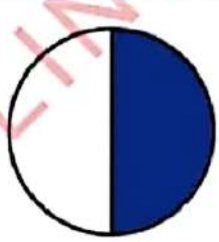
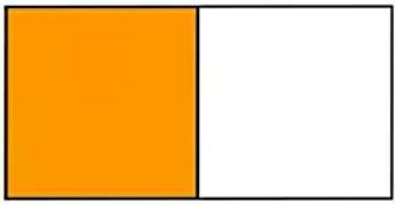
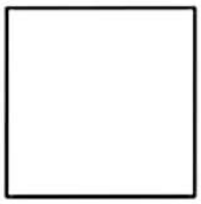
Whole object	First half	Second half
a tomato	half a tomato	half a tomato

Writing a half

In numeral, a half is written as $\frac{1}{2}$. It is read as a half or one over two.

Exercise 1

Write a half or $\frac{1}{2}$ in the blank spaces. Question 1 is an example.

1.		a half	$\frac{1}{2}$
2.		 a half	_____

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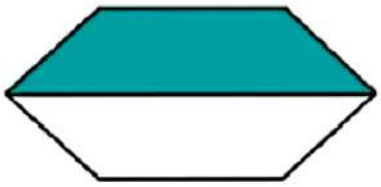
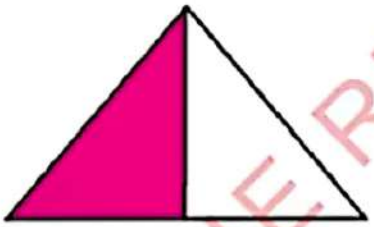
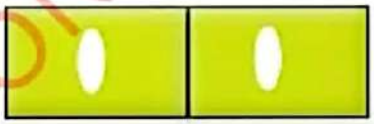
3.		_____	$\frac{1}{2}$
4.		a half	_____
5.		_____	$\frac{1}{2}$
6.		a half	_____
7.		_____	$\frac{1}{2}$

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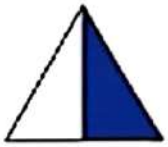
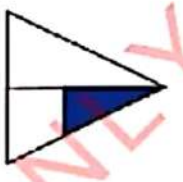
Exercise 2

Write Yes if the shaded part is a half. Write No if the shaded part is not a half.

1. 

4.
---

7. 

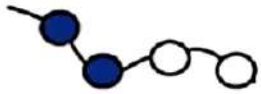
2.

5.

8.

3.

6.
---

9. 

Recognizing a quarter of a whole object

Four equal parts of an object form one whole object. A quarter is one of the four equal parts. For example, an orange can be cut into four equal pieces. Each piece is a quarter of the orange.

In numerals, a quarter is written as $\frac{1}{4}$. It is read as, a quarter or one over four.

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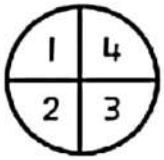
Example 1



Whole object

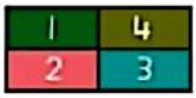
First quarter

Second quarter

Third quarter

Fourth quarter

Example 2



Whole object

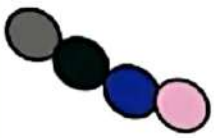
First quarter

Second quarter

Third quarter

Fourth quarter

Example 3



Whole object

First quarter

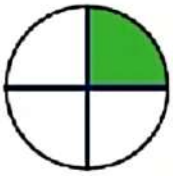
Second quarter

Third quarter

Fourth quarter

Exercise 3

Write 'a quarter' or $\frac{1}{4}$ in the given blank spaces.

1.		a quarter	___

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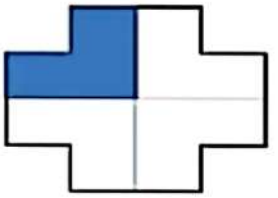
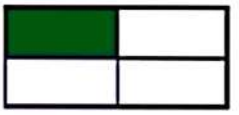
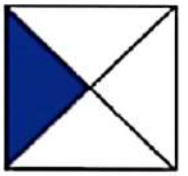
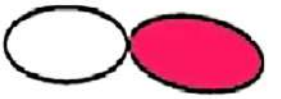
2.		____	$\frac{1}{4}$
3.		a quarter	____
4.		a quarter	____
5.		____	$\frac{1}{4}$

Exercise 4

Write $\frac{1}{2}$ or $\frac{1}{4}$ in the blank spaces.

1.
.....

2.
.....

3. 
.....

4.
.....

5.
.....

6.
.....

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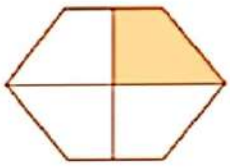
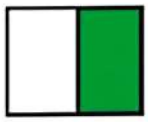
7. 

.....

8.

.....

9. 

.....

Recognizing one-third and two-thirds of a whole object

Three equal parts of the same object form one whole object. One-third is one of the three equal parts. For example, an orange can be cut into three equal pieces. Each piece is one-third of the orange. After removing one-third from an orange, the remaining part is two-thirds.

In numeral, one-third is written as $\frac{1}{3}$. It is read as, one-third or one over three. Two-thirds is written as $\frac{2}{3}$. It is read as, two-thirds or two over three.

The following pictures show an orange being divided into three equal parts.

An orange

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<table>
<tr><td> </td><td> </td></tr>
<tr><td colspan="2">The orange is divided into three equal parts.</td></tr>
<tr><td>
One piece of the orange is removed.</td><td>The piece of the orange removed is one-third.

One-third, $\frac{2}{3}$.</td><td>The remaining two pieces of the orange are two-thirds.

Two-thirds, $\frac{2}{3}$.</td></tr>
</table>

Activity

Let us play a one-third and two-thirds game.

We are three pupils. We are playing a one-third and two-thirds game. We stand in a straight line. One of us in the middle holds hands with the rest. Now, we are one whole object as the picture shows.

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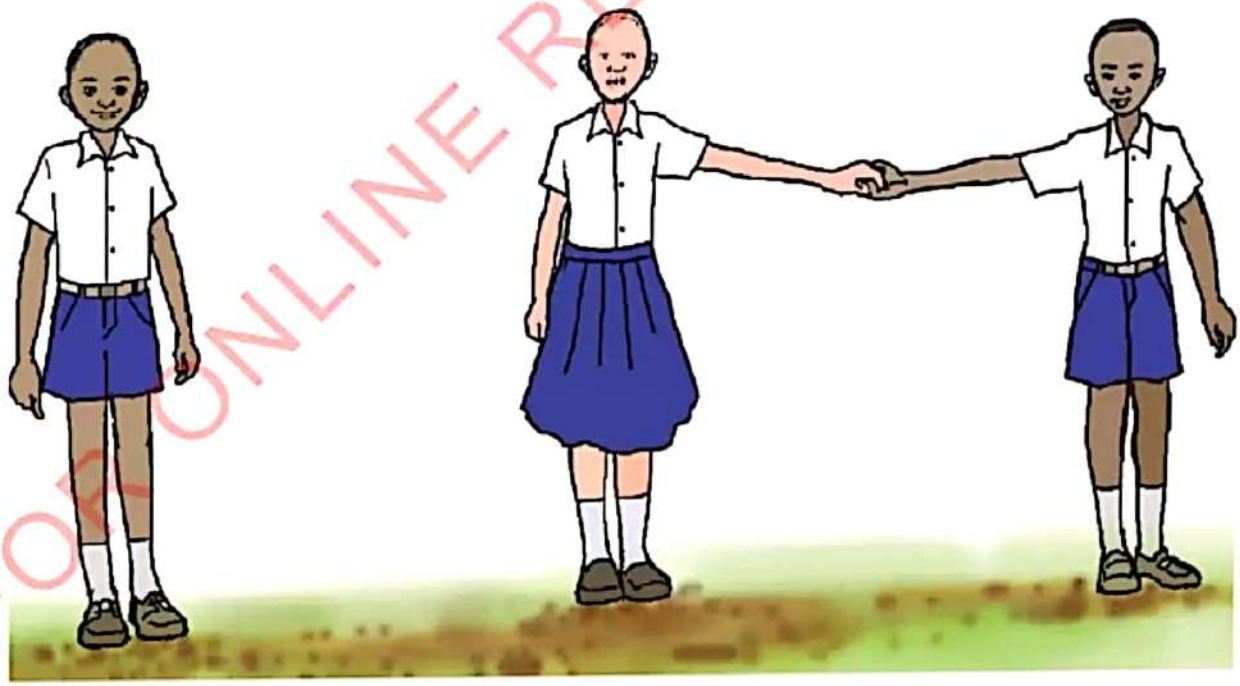
I have to leave the line. Goodbye. Look, now I am far from you. The following picture shows I am far from you.



I am one third of the three of us. What about you? We are two-thirds of the three of us.

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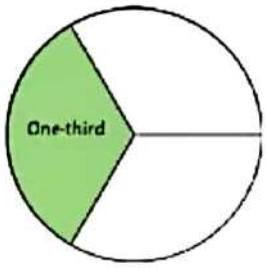
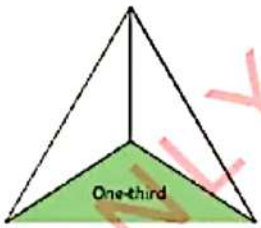
Example 1

Recognize one-third by looking at the shaded region.

1.

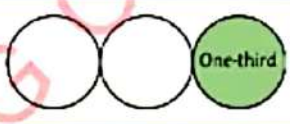


3.



2.

4.



Example 2

Recognize two-thirds by studying the shaded regions.

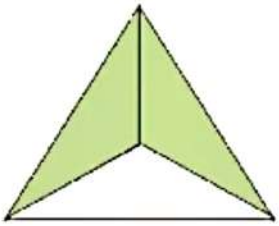
1.

Two-thirds

2.

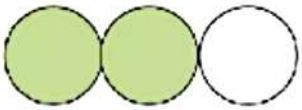
Two-thirds

3.



Two-thirds

4.



Two-thirds

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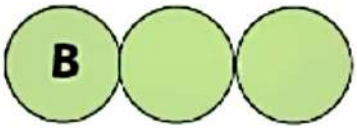
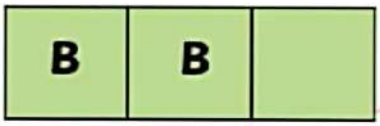
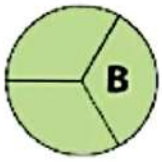
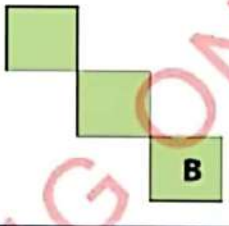
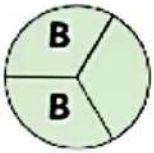
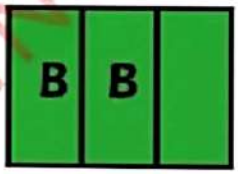
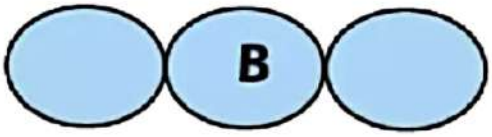
Exercise 5

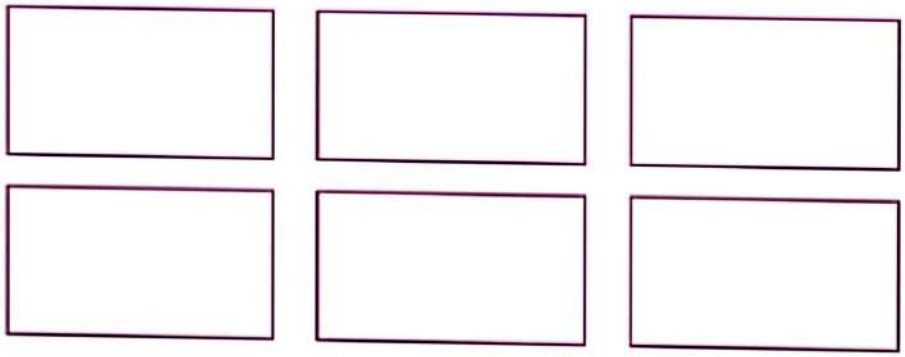
Write in words the name of the fraction labelled B.

1.		5.	
2.		6.	
3.		7.	
4.		8.	

Exercise 6

Answer the following questions:

1. Write one-third in numerals in each of the following boxes:



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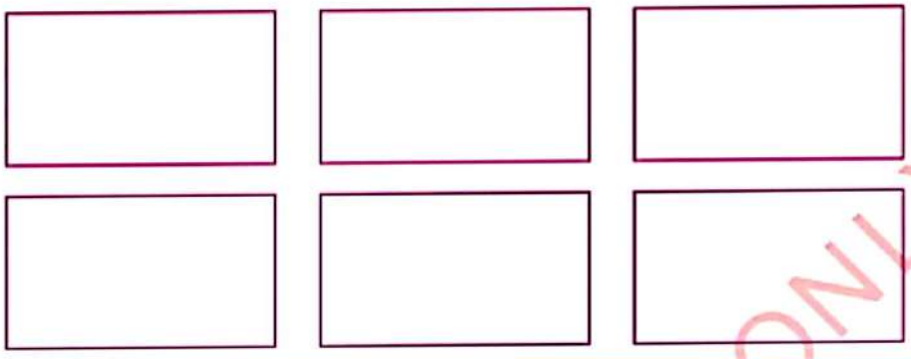
2. Write two-thirds in numerals in each of the following boxes:



Exercise 7

Write in words the fraction of the objects labelled A.

	Objects	Group	Name the fraction of the object labelled A.
1.		One fruit out of the three	One-third
2.			
3.			
4.			
5.			

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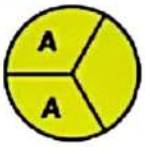
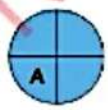
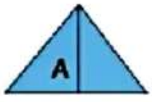
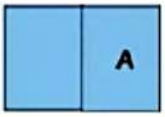
Exercise 8

Write in words and in numerals the fraction of the region labelled A.

	Figure	Name of the fraction labelled A in words	Name of the fraction labelled A in numerals
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			
9.			

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Exercise 9

Fill in the blanks.

	Fraction in numerals	Fraction in words
1.	$\frac{1}{2}$	
2.		one-third
3.	$\frac{1}{4}$	
4.	1	
5.		half
6.		whole object
7.	$\frac{1}{3}$	
8.		two-thirds
9.	$\frac{2}{3}$	
10.		one-fourth
11.		one-third
12.		two-thirds

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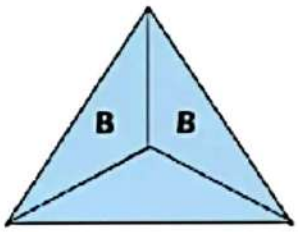
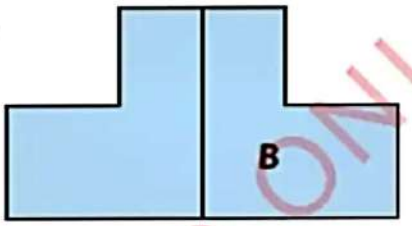
Exercise 10

Write in numerals the fraction labelled B in the following figures:

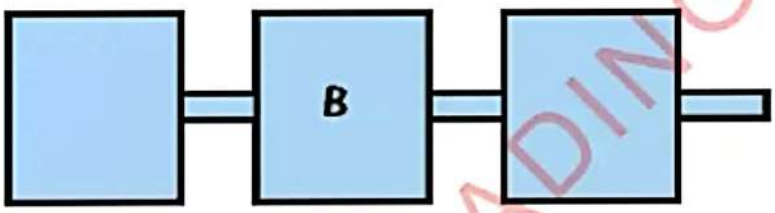
1.



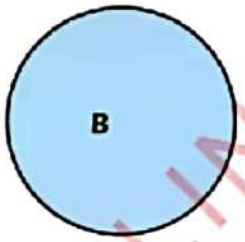
5.



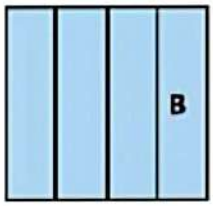
2.



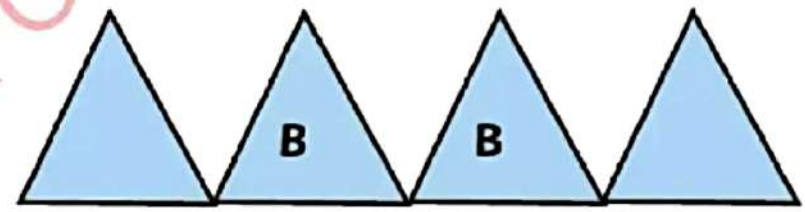
3.



6.



4.



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Recognizing fractions by using ICT

Fractions can be recognized by using ICT.

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Example

$\frac{1}{2}$	Solution Use the computer application in the TIE online library https:// tie.go.tz/ pages/download-software to divide numbers.

Steps

1. Open the application for recognizing fractions.
2. Read clearly the question that appears on the screen.
3. Write the answer by shading the part of the circle.
4. Click OK to check your answer.
5. Do other questions to practice more on recognizing fractions.

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Measurements

Distance

Distance is the length from one point to another. Objects can be near to or far from each other.

Study the following pictures.

Musa Ana

Tatu

1. Ana is near to Musa.
2. Ana is far from Tatu.

Exercise 1

Study the following pictures. Fill in the blanks with the words near to or far from.

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1. The ball is _____ the flask bottle.
2. The ball is _____ the chair.
3. The chair is _____ the ball.
4. The chair is _____ the flask bottle.

Exercise 2

Study the following pictures then answer the questions that follow.

1. Name the object that is near to the book.
2. Which object is far from the cup?
3. Which object is far from the book among the pen and the cup?
4. Which object is far from the pen?
5. Which object is near the cup?

Exercise 3

Study the following picture and answer the questions that follow.

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Questions

1. Which one among the post office and the open market is nearer to the school?
2. Which one among the post office and the black car on the road is nearer to the school?
3. Which one among the open market and the school is nearer to the post office?
4. Which one among the post office and the school is far from the open market?
5. Which one among the open market and the white car on the road is nearer to the post office?

Length

Length is a measure of horizontal or vertical distance. In our environment, there are long and short objects. Some objects have the same length. The vertical distance is called height.

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Example

The door is higher than the chair.

Recognizing standard measuring tools of length

Tape measure

Ruler

Recognizing non-standard tools for measuring length

footsteps	

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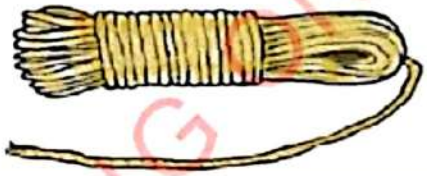
arm span	
rope	

Exercise 4

Write the name of the shortest object or person.

1.

Ana

Juma

2.

A cup

A flask bottle

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3.

A barrel

A bucket

Recognizing objects with equal lengths or heights

Example

Study the following pictures.

	The bottles have equal height
	The pencils have equal length
	The walls of the houses have equal height

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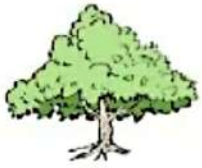
Exercise 5

Circle the object with a shorter length or height than the other.

1.	a	b
2.	a	b
3.	a	b
4.	a	b
5.	 a	b

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Activity

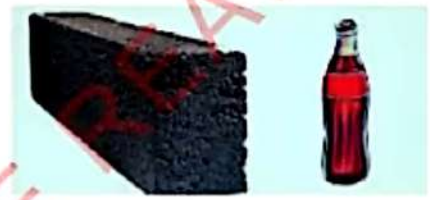
1. List the objects with equal lengths at your school.
2. List the objects with equal lengths from your home.
3. Name the objects with equal lengths that you see on the way.

Mass

Mass is the measure of heaviness of an object. In our environment, there are heavier and lighter objects.

Example

A brick is heavier than a bottle of soda.



Exercise 6

In each group, write the name of the object which is heavier than the other.

1.	car motorcycle

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2.	hoe spoon
3.	rat cat

Recognizing standard tools for measuring mass

Beam balance	

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Recognizing non-standard tools for measuring mass

The following are some non-standard tools for measuring mass:

sacks	
buckets	
tin	
hands	

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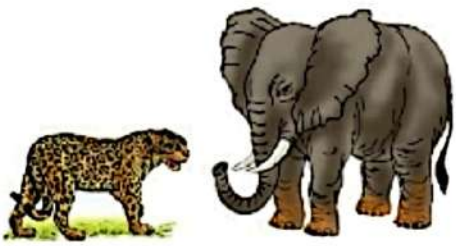
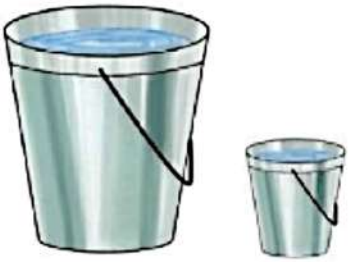
Exercise 7

In each group, circle the object which is heavier than the other.

1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	

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Volume

Volume is the space that can be occupied by an object. The volume of objects can be compared by looking at the amount of space that fills the device. There are objects with large volumes and others with small volumes.

Example

A cup has a smaller volume than a bucket.

Recognizing standard tools for measuring volume

Measuring cylinder

beaker

bucket

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Recognizing non-standard tools for measuring volume

bottle

jug

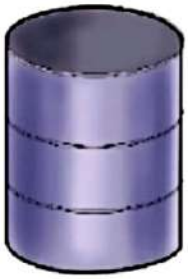
glass

Recognizing units for measuring volume

Some of the units used to measure volume are millilitres and litres.

Exercise 8

In each group, name the object whose volume is smaller than that of the other.

1.	 burrel bucket
2.	jug glass

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3.	bowl	pot

Exercise 9

1. Which bottle has more soda than the other?

(a) (b)

2. Which jug has more juice than the other?

(a) (b)

3. Which objects at your home have the same volume?
4. Which objects at your school have the same volume?
5. Which objects in the shop have the same volume?
6. Mention containers with equal volumes for storing water at your home.

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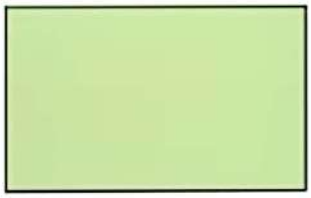
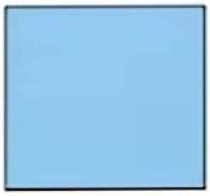
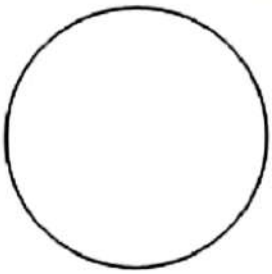
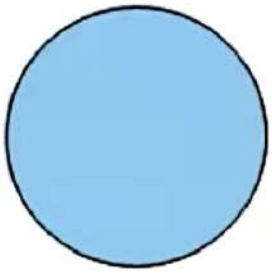
Shapes

Plane figures

Plane figures are shapes drawn on flat surfaces.

Example

The following are examples of plane figures and their shapes.

1.	rectangle	 The shape of a rectangle
2.	square	 The shape of a square
3.	 circle	 A shape of a circle

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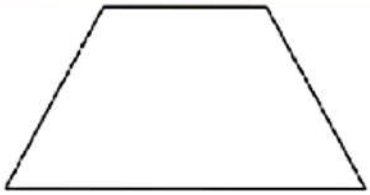
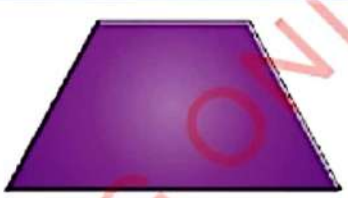
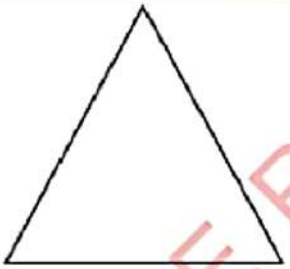
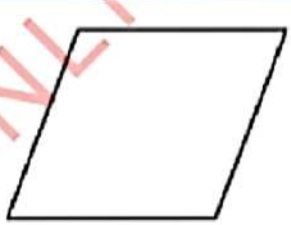
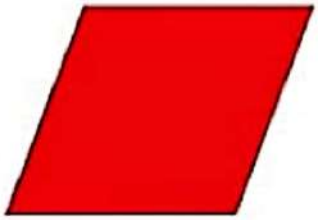
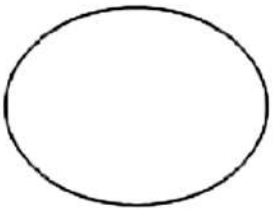
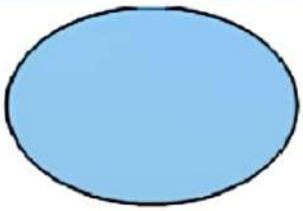
4.	parallelogram	The shape of a parellelogram
5.	 trapezium	 The shape of a trapezium
6.	 triangle	The shape of a triangle
7.	 rhombus	 The shape of a rhombus
8.	 oval	 The shape of an oval

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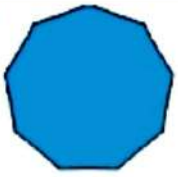
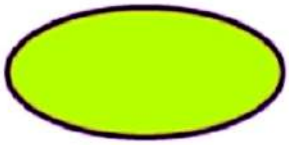
Exercise 1

1. Name some objects that have the shape of a circle.
2. Which objects in your classroom have the rectangular shape?
3. Name some objects that have the shape of a triangle.
4. Mention some objects with the square shape.
5. Which objects have the oval shape?
6. List the names of all four-sided figures from your answer.

Exercise 2

Study the following figures and answer the questions that follow.

a.		b.		c.	
d.		e.		f.	
g.		h.		i.	

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1. List all similar shapes by writing their letters together.
2. Write the letters of all the shapes of triangles.
3. Write the letters of all the shapes with four sides.
4. Write the names of all the shapes of plane figures you know.

Activity

Playing ukuti-ukuti

Let us play together by joining our hands.

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Let us sing the song and then answer the questions that follow.

Leader: Ukuti-ukuti.
Followers: Wa mnazi wa mnazi.
Leader: Mwenzetu, mwenzetu.
Followers: Kagongwa, kagongwa.
Leader: Na nini, na nini?
Followers: Na gari, na gari.
Leader: Tumpeleke hospitalini asije kusema kwa baba yake.
Followers: Yesa, yesa, yesa, yee.
Leader: Cheza kidogo.
Followers: Yesa, yesa, yesa yee.

Questions

1. What shape does the arrangement of players relate to?
2. What other game is similar to the game you played?
3. Which game has a rectangular arrangement?

Exercise 3

1. Draw a circle by tracing a 100 or 200 shillings coin.
2. Draw a triangle by tracing the shape of a triangular object.

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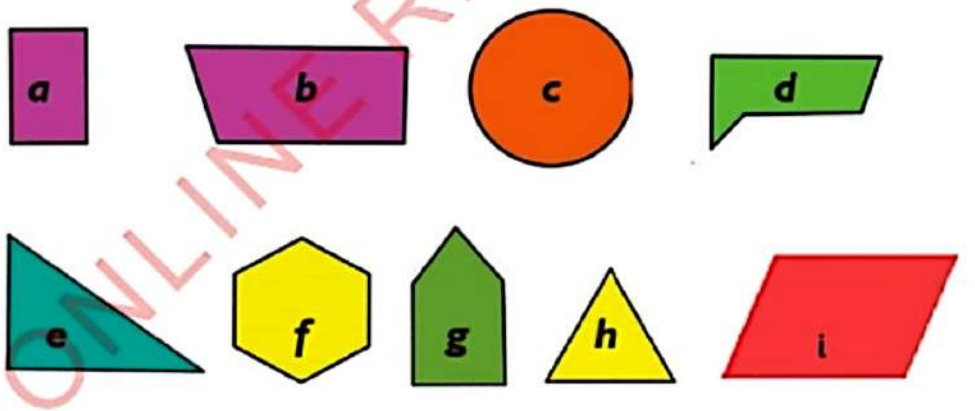
3. Draw a rectangle by tracing a matchbox.
4. Draw an oval by tracing the shape of an oval object.
5 Draw a square by tracing the shape of a square object.
6. Draw a parallelogram by tracing the shape of a parallelogram object.
7. Draw a trapezium by tracing the shape of trapezium object.

Exercise 4

Study the following figures and then answer the questions that follow.



Questions

1. Write the letters of all four sided shapes.
2. Write the letter appearing in the shape of a circle.
3. Write the letters of all the shapes of triangles.

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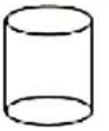
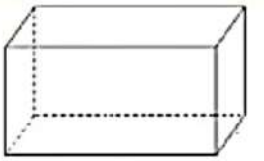
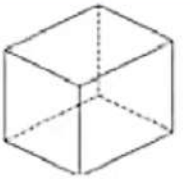
4. Mention the shapes that are different from the shape of a triangle or a circle.
5. Recognizing non-plane figures
6. Non-plane figures have capacities or volumes.

The following are examples of non-plane figures and their real objects.

Non-plane figures	Real objects	Figures
		 cylinder
		cone
		 cuboid
		 cube

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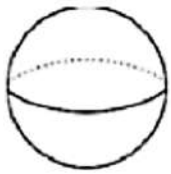
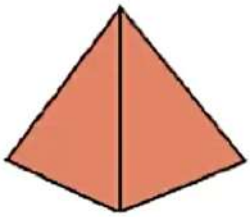
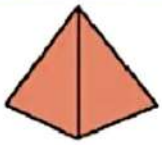
		 sphere
		pyramid

Exercise 5

Identify and then draw the plane figures in the following shapes.

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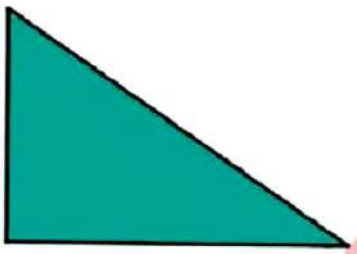
Exercise 6

Write plane or non-plane in the blanks.

1.

This shape is _____

2. 

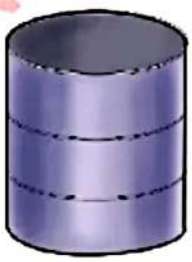
This shape is _____

3.

This shape is _____

4.

This shape is _____

5. 

This shape is _____

6.

This shape is _____

7. Draw any two plane shapes.

8. Draw any two non-plane shapes.

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Exercise 7

Draw and colour the following shapes:

1. A shape of a rectangle	2. A shape of a parallelogram
3. A shape of a triangle	4. A shape of a trapezium
5. A shape of a square	6. A shape of a rhombus
7. A shape of a circle	8. A shape of an oval
9. A shape of a sphere	10. A shape of a pyramid

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11. A shape of a cone	12. A shape of a cylinder
13. A shape of a cuboid	14. A shape of a cube

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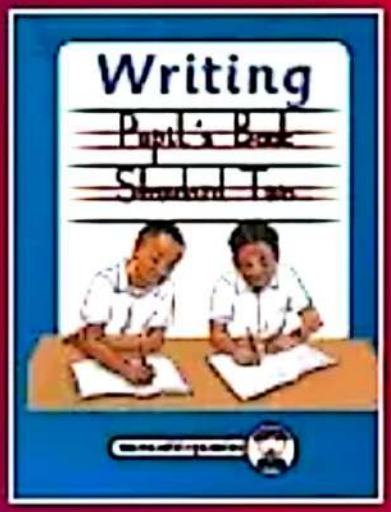
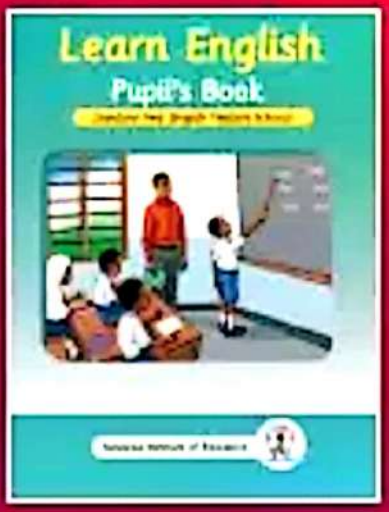
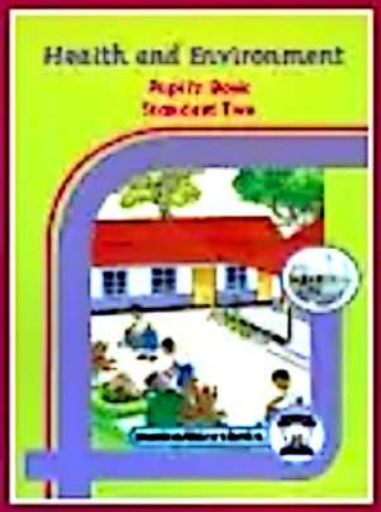
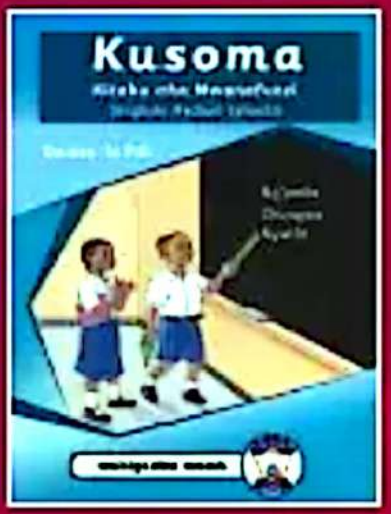
Standard Two









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